

SYSTEMATIC SEARCH FOR SINGULARITIES

SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D EULER
FLOWS BASED ON THE VOIGT REGULARIZATION

BY

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TITLE: SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D
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Lay Abstract

A lay abstract of not more 150 words must be included explaining the key goals and contributions of the thesis in lay terms that is accessible to the general public.

The Navier-Stokes equations model the behaviour of viscous incompressible fluids. The question of whether or not the Navier-Stokes equations, or the inviscid Euler equations, are a valid description of fluid mechanics remains unanswered. A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. If the equations form a singularity from smooth initial conditions, then they are not valid descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to not form singularities from smooth initial conditions. However, by decreasing the regularization parameter we approximate Euler flows. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify smooth initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

Abstract

Abstract here (no more than 300 words). An abstract of not more than 300 words must be included and indicates the major emphasis of the thesis, new discoveries, and its contribution to knowledge. The style of abstract varies from discipline to discipline but the student should follow an appropriate style. This page must be numbered 'iv'.

The Navier-Stokes equations are a system of partial differential equations used to model the behaviour of viscous incompressible fluids. The question of whether or not unique classical solutions of the Navier-Stokes equations, and inviscid Euler equations, exist for all time, given smooth initial conditions, remains unanswered for the three-dimensional problem.

A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. The formation of a singularity from smooth initial conditions would invalidate the equations as accurate descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to be globally well-posed from smooth initial conditions. However, by decreasing the regularization parameter the solutions of the Euler-Voigt equations converge

to those of the Euler equations. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify smooth initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

Your Dedication
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Acknowledgements

Acknowledgements go here. An expression of thanks for assistance given by the Supervisor of research and by others should be either set forth on a separate page or incorporated into the Preface (if there is one). These and all subsequent preliminary pages listed under (f), (g) and (h) must be numbered in lower case Roman numerals, i.e., ‘iv’, ‘v’, ‘vi’ etc.

Dr. Bartosz Protas, Dr. Xinyu Zhao, Dr. Jörn Davidsen, members of committee,
Parents

I would like to thank Dr. Bartosz Protas for his support and guidance on my research project. I also want to thank Dr. Jörn Davidsen of the Complexity Science Group for supervising my undergraduate research and inspiring me to pursue computational physics.

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Notation, and Abbreviations

Notation

x	Plain-text to denote scalar x
$\boldsymbol{x}$	Bold-text to denote vector x
$\boldsymbol{x}^\top$	Transpose of vector x
I	Identity Operator
∇	Gradient Operator
Δ	Laplacian Operator
$:=$	Equal to by definition

Abbreviations

PDEs	Partial Differential Equations
3D	three-dimensional or three dimensions
TGV	Taylor Green Vortex

Declaration of Academic Achievement

Dr. Zhao developed most of the mathematics for the subspace V (2.26), along with Theorem 2.1.3 and its proof. Dr. Zhao also developed the mathematics for much of the Euler-Voigt optimization problem described in Chapter 3, including the objective functional (3.32), the linearization of the Euler-Voigt equations (3.37), the adjoint system (3.38), and the gradient of the objective functional (3.44). The Riemannian conjugate gradient method for the Euler-Voigt optimization problem in Section 3.4 was developed by Dr. Zhao and Dr. Protas. This includes the tangent space (3.45), the projection operator (3.48), the search direction (3.52), and step size (3.55). Dr. Zhao also provided the data files for initial guesses () and () .

The retraction operator (3.49), and the vector transport operation (3.50) were originally developed by Dr. Zhao and Dr. Protas but I modified these formulations to suit the present problem. The formulation of the manifold $\mathcal{M}_C$ (2.28) was based on work by Dr. Zhao, but the calculation of the constraint parameter C was my work. This included the calculations of the $\dot{H}^1$ seminorm (2.30) and vorticity (2.31) of the general Taylor Green Vortex, which are included in Appendices E and F. The results in Appendices B, G, H, and I are my re-derivations of Dr. Zhao's original results. The derivation in Appendices C and D are my re-derivations of the well known Euler and

Euler-Voigt scaling properties.

For the numerical models, Dr. Zhao created the original code base which I modified to suit this project. The determination of the numerical model constants, the time and spatial resolutions, regularization parameter α , and Gevrey parameter σ were also my work. These values were validated through analysis of the stability and accuracy of the model, conservation of the α -energy, comparison of the effect of different values of σ , examination of the solutions' Fourier spectrum, and the kappa test, all of which were my work and are described in Chapter 4. I also validated the results of the numerical models through comparison with the relevant findings of [5] and [20]. I performed all of the computations to obtain the final numerical results. The findings and conclusions in Chapters 5 and 6 are my work. The Python code used to perform the data analysis and produce the graphs is also my own work.

1 Introduction

1.1 Singularity Formation in Models of 3D Fluid Flows

The three-dimensional (3D) Navier-Stokes equations are a set of partial differential equations (PDEs) that describe the motion of homogeneous, viscous, incompressible fluids. The Navier-Stokes equations on the domain $\Omega \subseteq \mathbb{R}^3$ are defined as

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = 0 \quad \text{in } \Omega \times (0, T], \quad (1.1a)$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega \times [0, T], \quad (1.1b)$$

$$\mathbf{u}(\mathbf{x}, 0) = \boldsymbol{\eta} \quad \text{in } \Omega \quad (1.1c)$$

where $\mathbf{u} = [u_1, u_2, u_3]^\top$ is the fluid velocity field, p is the scalar pressure, $\nu > 0$ is the coefficient of kinematic viscosity, and T is the length of the time interval. The field $\boldsymbol{\eta}$ is the initial condition assumed to be divergence-free. Finally, the gradient operator $\nabla = [\partial_1, \partial_2, \partial_3]^\top$ is used to define the gradient of the pressure, the tensor $[\nabla \mathbf{u}]_{ij} = \partial_j u_i$ for $i, j = 1, 2, 3$, the divergence of the velocity field, and the Laplacian operator $\Delta \mathbf{u} = \nabla \cdot (\nabla \mathbf{u})$. Because of their generality, the Navier-Stokes equations are utilized to model the behaviour of systems in fields as disparate as aero/hydrodynamics, weather prediction, biophysics, and astrophysics [10, 12]. Despite the numerous fields

that make use of the 3D Navier-Stokes equations, and extensive research examining their properties, it is only known that weak Leray-Hopf solutions exist globally in time [22]. However, these solutions may contain singularities and their uniqueness remains unproven in most cases. It is also yet to be proven whether or not smooth classical solutions, defined in the pointwise sense, exist globally in time [28]. In fact, this is the basis of one of the seven Millennium Problems posed by the Clay Mathematics Institute which offers a substantial monetary prize for the confirmation of the existence, or non-existence, of a classical solution corresponding to smooth initial conditions [14]. The space of smooth functions and all other function spaces utilized in this analysis are defined in Appendix A. It is common for scientific research that focuses the fundamental properties of the Navier-Stokes equations to utilize either the unbounded domain $\Omega := \mathbb{R}^3$, the ~~axisymmetric~~ domain $\Omega := \{(r, z) : r < 1, z \in \mathbb{T}\}$, or the periodic 3D torus $\Omega := \mathbb{T}_L^3$ with domain size $L > 0$ [28]. For this analysis, we use the 3D torus of unit size ($L = 1$) with ~~periodic conditions on the boundary $\partial\Omega$.~~

A central question in fluid mechanics is whether or not the Navier-Stokes equations are globally well-posed or if singularities can form from smooth initial conditions. A singularity occurs when the derivative of the velocity field becomes infinite at a single point [25]. The formation of a singularity in a fluid flow, from smooth initial conditions, would invalidate the Navier-Stokes equations as an accurate description of fluid flows [28]. The question about singularity formation from smooth initial conditions also remains open for the Euler equations [16]. The Euler equations are the inviscid

form of the Navier-Stokes equations, obtained by setting $\nu = 0$

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p = 0 \quad \text{in } \Omega \times (0, T], \quad (1.2a)$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega \times (0, T], \quad (1.2b)$$

$$\mathbf{u}(\mathbf{x}, 0) = \boldsymbol{\eta} \quad \text{in } \Omega. \quad (1.2c)$$

The Euler equations are often studied because they are simpler to analyse than the Navier-Stokes equations due to the absence of the viscous dissipation. The local existence of classical solutions to the Euler equations was proven in [19] for solutions defined in Sobolev spaces.

Theorem 1.1.1 ([19]). *If $\boldsymbol{\eta} \in H^m(\mathbb{T}^3)$ for some $m > 5/2$ and satisfies $\nabla \cdot \boldsymbol{\eta} = 0$, then there exists a time $T = T(\|\boldsymbol{\eta}\|_{H^m}) > 0$ such that (1.2) has a unique solution $\mathbf{u}(\cdot; \boldsymbol{\eta}) \in C([0, T]; H^m) \cap C^1([0, T]; H^{m-1})$,*

where $\|\cdot\|_{H^m}$ is the standard Sobolev norm, which is described in Appendix A. If a singularity is to form in the solutions to the Euler equations, it must occur after the minimum local existence time defined in Theorem 1.1.1. There are many different conditional regularity results for determining if a singularity has formed, the best known of which is the Beale-Kato-Majda (BKM) criterion [3] which asserts

Theorem 1.1.2 ([3]). *A smooth solution $\mathbf{u}$ of the Euler system develops a singularity at $t = t_0$ if and only if*

$$\lim_{t \rightarrow t_0} \int_0^t \|\boldsymbol{\omega}(\tau)\|_{L^\infty} d\tau = \infty, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u}, \quad (1.3)$$

where $\|\cdot\|_{L^\infty}$ is the standard Lebesgue L^∞ norm, which is described in Appendix A.

A key difference between the Navier-Stokes and Euler equations is that there is more numerical evidence that indicates singularity formation in Euler is possible from smooth initial conditions. For example, [4, 5] presented numerical evidence for singularity formation in the Euler equations on a 3D torus ($L = 2\pi$). Hou et al. produced several papers which provide numerical evidence for singularity formation in the 3D axisymmetric Euler equations [24, 18, 17] and which proved singularity formation in the 3D axisymmetric Euler equations with rigid boundaries and periodic boundary conditions in the axial direction [8, 9]. We make special note of the optimization-based approach of [30], which utilized a Riemannian conjugate gradient method to identify 'extreme' initial conditions, and reports numerical results consistent with singularity formation in Euler flows on the 3D unit torus. Finally, [20] which provides numerical evidence for singularity formation in the Euler equations based on the Voigt regularization. These final two papers are of special significance for us because they form the primary basis for the motivation and methodology of this analysis.

1.2 3D Euler-Voigt Equations

The Euler-Voigt equations are an inviscid regularization of the Euler equations (1.2) which have been shown to be globally well-posed and so do not form singularities from smooth initial conditions [7]. The Euler-Voigt equations are defined as

$$-\alpha^2 \partial_t \Delta \mathbf{u}_\alpha + \partial_t \mathbf{u}_\alpha + \mathbf{u}_\alpha \cdot \nabla \mathbf{u}_\alpha + \nabla p_\alpha = 0 \quad \text{in } \Omega \times (0, T], \quad (1.4a)$$

$$\nabla \cdot \mathbf{u}_\alpha = 0 \quad \text{in } \Omega \times (0, T], \quad (1.4b)$$

$$\mathbf{u}_\alpha(\mathbf{x}, 0) = \boldsymbol{\eta}_\alpha \quad \text{in } \Omega, \quad (1.4c)$$

where the regularization is accomplished with the addition of the term $-\alpha^2 \partial_t \Delta \mathbf{u}_\alpha$ in which $\alpha > 0$ is the regularization parameter with units of length. This term stabilizes the system without dissipation, which would remove energy from the system [20]. The regularization term acts as a spectral filter and suppresses the development of small scale structures with length-scale smaller than α [26]. We introduce the subscript α to distinguish between solutions of (1.2) and (1.4) i.e., $\mathbf{u}$ and $\mathbf{u}_\alpha$.

For (1.2) the total kinetic energy is conserved so long as the solution remains well-posed in the classical sense

$$E_k(t) := \frac{1}{2} \|\mathbf{u}(t)\|_{L^2}^2 = E_k(0), \quad (1.5)$$

where the norm $\|\cdot\|_{L^2}$ here is the standard Lebesgue L^2 norm. For (1.4) the conserved quantity is known to be the α -energy. The α -energy is defined as

$$E_\alpha(t) := \|\mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0), \quad (1.6)$$

where $\|\mathbf{u}_\alpha(t)\|_{L^2}^2$ is the energy term, and $\|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2$ is the enstrophy term, scaled by α^2 .

1.3 Outline of the Thesis

The goal of this analysis is to search for possible singularities in Euler flows corresponding to smooth initial conditions. This search will be performed based on a PDE-constrained optimization problem in which we will identify initial condition that produce the most 'extreme' behaviour in Euler-Voigt flows. In Chapter 2 we present

the properties of the Euler-Voigt equations, including the blow-up criteria, scaling property, and manifold over which we search for initial conditions. In Chapter 3 we present the formulation of the optimization problem. This includes the description of our objective functional, the gradient calculation, and the Riemannian conjugate gradient method used to compute our search direction and iterative relation. In Chapter 4 we describe the numerical methods used to perform the calculations, including approximation in space and time, Gevrey parameters, and the regularization parameter. In addition, we validate the computations using the Fourier spectrum, kappa test, and conservation of α -energy. In Chapter 5 we present our results. We begin with the solutions of the Euler-Voigt equations ~~corresponding to initial guesses,~~ without optimization, as $\alpha \rightarrow 0$. These results indicate that singularities do not form in Euler flows corresponding to these initial guesses, within the time window. We then present our optimization results as $\alpha \rightarrow 0$, which indicate that RESULTS RESULTS RESULTS. Finally, in Chapter 6 we summarize our findings and present our conclusion.

2 Properties of the Euler-Voigt Equations

In this chapter we prove **a theorem** which provides the criterion used to determine if a singularity forms in Euler flows from smooth initial conditions. Additionally, we construct **a manifold over which** we search for optimal smooth initial conditions. Finally, we examine the structure of the initial **guesses** including the Taylor Green Vortex, **and two optimized initial conditions from the results of [30]**. Together, these form the basis for our optimization problem.

2.1 Finite Time Blow-up Criterion

In order to begin our search for finite-time singularity formation, we must first describe the criteria used to detect singularity formation in Euler flows. In **[21]** a new criterion for identifying singularity formation in Euler flows was developed which utilizes the Euler-Voigt equations. This criterion relies on the convergence of the solutions of **(1.4)** to the solutions of **(1.2)** as $\alpha \rightarrow 0$.

Theorem 2.1.1 (**[21]**). *Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$, let $\boldsymbol{u}, \boldsymbol{u}_\alpha \in C([0, T], H^s(\Omega)) \cap C^1([0, T], H^{s-1}(\Omega))$ be the corresponding solutions to **(1.2)** and **(1.4)**, respectively, where $0 < T < T_{max}$ and T_{max} is the maximal time for which*

a solution to (1.2) exists and is unique. For all $t \in [0, T]$, we have

$$\begin{aligned} & \|\mathbf{u}(t) - \mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla(\mathbf{u}(t) - \mathbf{u}_\alpha(t))\|_{L^2}^2 \leq \\ & (\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2) e^{Ct} + C\alpha^2(e^{Ct} - 1). \end{aligned} \quad (2.7)$$

Based on this theorem we know that if

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \rightarrow 0, \quad \text{as } \alpha \rightarrow 0, \quad (2.8)$$

then for all $t \in [0, T]$,

$$\|\mathbf{u}(t) - \mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla(\mathbf{u}(t) - \mathbf{u}_\alpha(t))\|_{L^2}^2 \rightarrow 0. \quad (2.9)$$

Therefore, solutions of the Euler-Voigt system (1.4) converge to the corresponding Euler (1.2) flows in H^1 uniformly in time as $\alpha \rightarrow 0$. Using this result, the following theorem was proved in [20].

Theorem 2.1.2. *Let $\boldsymbol{\eta}_\alpha \in H^s$, $s \geq 3$, with $\nabla \cdot \boldsymbol{\eta}_\alpha = 0$ and let $\mathbf{u}_\alpha$ be the corresponding unique solution of (1.4). Suppose that there exists a time $T^* > 0$ such that*

$$\limsup_{\alpha \rightarrow 0} \left(\alpha \sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \right) > 0. \quad (2.10)$$

Then, the corresponding solution of the Euler equations (1.2) with the initial condition $\boldsymbol{\eta}_\alpha$ must develop a singularity within the interval $[0, T^]$.*

We note that this theorem relies on the solutions of (1.2) and (1.4) starting from the same initial condition, i.e., $\boldsymbol{\eta} = \boldsymbol{\eta}_\alpha$. This necessarily satisfies (2.8) as $\alpha \rightarrow 0$. However, in order to use this result in our study, we need to generalize Theorem 2.1.2

for the case where (2.8) is satisfied but $\boldsymbol{\eta} \neq \boldsymbol{\eta}_\alpha$. Therefore, we modify Theorem 5.2 from [21] as follows.

Theorem 2.1.3 (X. Zhao, private communication). *Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$ satisfying*

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \rightarrow 0, \quad \text{as } \alpha \rightarrow 0, \quad (2.11)$$

let $\mathbf{u}, \mathbf{u}_\alpha \in C([0, T], H^s(\mathbb{T}^3)) \cap C^1([0, T], H^{s-1}(\mathbb{T}^3))$ be the corresponding solutions to (1.2) and (1.4), respectively. If there exists a finite time $T^ > 0$ such that the solutions $\mathbf{u}_\alpha$ satisfy*

$$\limsup_{\alpha \rightarrow 0} \left(\alpha^2 \sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 \right) > 0, \quad (2.12)$$

then $\mathbf{u}$ develops a singularity in the interval $[0, T^]$.*

Proof. Suppose that the solution $\mathbf{u}$ of the Euler equations (1.2) with the initial condition $\boldsymbol{\eta}$ is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have, cf. (1.6)

$$\|\mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \boldsymbol{\eta}_\alpha\|_{L^2}^2, \quad (2.13)$$

thus

$$\alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \boldsymbol{\eta}_\alpha\|_{L^2}^2 - \|\mathbf{u}_\alpha(t)\|_{L^2}^2. \quad (2.14)$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$\sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2}^2. \quad (2.15)$$

Taking the limsup as $\alpha \rightarrow 0$, we obtain

$$\limsup_{\alpha \rightarrow 0} \left(\sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 \right) = \limsup_{\alpha \rightarrow 0} \left(\|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \boldsymbol{\eta}_\alpha\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2}^2 \right). \quad (2.16)$$

We now make use of Theorem 2.1.1, which demonstrates that the convergence of the solutions of the Euler-Voigt equations (1.4) to the solution of the Euler equations (1.2) is uniform on $[0, T^*]$. Therefore we can reduce the above equation to

$$\limsup_{\alpha \rightarrow 0} \left(\sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 \right) = \|\boldsymbol{\eta}\|_{L^2}^2 - \|\mathbf{u}(t)\|_{L^2}^2. \quad (2.17)$$

Since the energy of regular solutions of the Euler equations is conserved, we have

$$\limsup_{\alpha \rightarrow 0} \left(\sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 \right) = 0, \quad (2.18)$$

which contradicts the hypothesis in the theorem that the above expression is positive.

Equality (2.17) holds because

$$\begin{aligned} \limsup_{\alpha \rightarrow 0} \left| \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2} - \|\boldsymbol{\eta}\|_{L^2} \right| &\leq \limsup_{\alpha \rightarrow 0} \|\mathbf{u}_\alpha(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\ &\leq \limsup_{\alpha \rightarrow 0} \|\mathbf{u}_\alpha(t_\alpha) - \mathbf{u}(t_\alpha)\|_{L^2}^2 + \|\mathbf{u}(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\ &= 0, \end{aligned}$$

where the reverse triangle and triangle inequalities were used and t_α is the time at which

$$\inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2} = \|\mathbf{u}_\alpha(t_\alpha)\|_{L^2}. \quad (2.20)$$

□

2.2 Initial Conditions

We have established Theorem 2.1.3 which we will use to identify singularity formation in Euler flows corresponding to smooth initial conditions. In order to formulate our optimization problem we must now describe the function space we perform our search in. Theorems 2.1.1, 2.1.2, and 2.1.3 presented in Section 2.1 require initial conditions $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$. However, the pseudo-spectral methods used in this study to numerically solve system (1.4) require the functions to be analytic in order to have exponential convergence [6]. For the Navier-Stokes equations (1.1), the viscous term acts to instantaneously regularize the solution in space. Therefore, if the initial condition has only Sobolev regularity, the resulting flow becomes real-analytic (with Gevrey regularity defined below) at $t = 0^+$. This regularity allows the solution to achieve exponential convergence.

The Euler equations (1.2) do not include any viscous effects and so the solution simply stays at the same level of smoothness as the initial condition, unless blow-up occurs. For Euler-Voigt the regularization term keeps the solution globally well-posed but does not improve the regularity [21]. Therefore, we must have analytic initial conditions to make use of the pseudo-spectral methods. For all the models considered here, a requirement is that the flow is divergence free at all times, including the initial conditions. Finally, on a periodic domain the quantity $\int_{\Omega} \boldsymbol{\eta}_\alpha d\mathbf{x}$ is an invariant of motion for solutions of the Euler-Voigt equations. Due to this, we assume without loss of generality that the initial conditions have a mean of zero. In order to identify initial conditions that satisfy these criteria, we first consider analytic initial conditions that

belong to the Gevrey class

$$G^\sigma := \left\{ \mathbf{v} \in H^s(\mathbb{T}^3) : \|\mathbf{v}\|_{G^\sigma} = \langle \mathbf{v}, \mathbf{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0, \quad (2.21)$$

with the norm defined as

$$\forall \mathbf{v}, \mathbf{u} \in G^\sigma, \quad \langle \mathbf{v}, \mathbf{u} \rangle_{G^\sigma} := \sum_{\mathbf{j} \in \mathbb{Z}^3} (1 + |2\pi \mathbf{j}|^2)^s e^{4\pi\sigma|\mathbf{j}|} \widehat{\mathbf{v}}_{\mathbf{j}} \cdot \overline{\widehat{\mathbf{u}}_{\mathbf{j}}} \quad (2.22a)$$

$$= \int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v} \cdot \mathbf{u} \, d\mathbf{x}, \quad (2.22b)$$

where hat denotes the Fourier coefficient

$$\widehat{\mathbf{u}}_k = \int_{\Omega} e^{-i\mathbf{k} \cdot \mathbf{x}} \mathbf{u}(\mathbf{x}) \, d\mathbf{x}, \quad (2.23)$$

overbar denotes complex conjugation, and the operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D|\mathbf{v}} \right]_{\mathbf{j}} := 2\pi|\mathbf{j}| \widehat{\mathbf{v}}_{\mathbf{j}}, \quad \left[\widehat{e^{\sigma|D|}\mathbf{v}} \right]_{\mathbf{j}} := e^{2\pi|\mathbf{j}|} \widehat{\mathbf{v}}_{\mathbf{j}}. \quad (2.24)$$

The parameter σ determines the relative 'size' of the Gevrey space

$$G^{\sigma_2} \subset G^{\sigma_1} \subset G^0 = H^s, \quad 0 < \sigma_1 < \sigma_2, \quad (2.25)$$

which is proved in Appendix B. The value of σ is chosen in Section 4.3. In order to satisfy the divergence free condition, we introduce the following subspace in which

optimal initial conditions will be sought

$$V := \{\mathbf{v} \in G^\sigma : \nabla \cdot \mathbf{v} = 0, \int_{\mathbb{T}^3} \mathbf{v} \, d\mathbf{x} = 0\}. \quad (2.26)$$

2.3 Scaling Symmetries of Euler and Euler-Voigt Systems

Both (1.2) and (1.4) possess a scaling property. Given a solution $\mathbf{u}$, or $\mathbf{u}_\alpha$, we can construct a new solution

$$\mathbf{v}(\mathbf{y}, \tau) = \frac{1}{\gamma} \mathbf{u}(\beta \mathbf{x}, \lambda t), \quad \mathbf{y} = \beta \mathbf{x}, \quad \tau = \lambda t \quad (2.27)$$

where $\lambda, \gamma, \beta > 0$ must satisfy $\beta/(\lambda\gamma) = 1$, which is proven in Appendices C and D, respectively for the Euler (1.2) and Euler-Voigt (1.4) equations. Our goal is to find a time interval $[0, T]$ that is sufficiently large so that it is outside the interval of local existence established by Kato's theorem 1.1.1 and so allows for the potential of singularity formation in Euler flows. Due to this scaling property of the Euler-Voigt equations (2.27), the **length of the time interval** will depend on the 'size' of our solution (γ) and the spatial scale (β). Our domain size will not change, and so in order to keep a consistent time scale we must fix γ between all initial conditions. To accomplish this, we restrict our discussion to initial conditions with a fixed $\dot{H}^1$ seminorm which belong to a closed manifold $\mathcal{M}_C \subset V$ defined as cf. (2.26)

$$\mathcal{M}_C = \{\mathbf{v} \in V : \|\mathbf{v}\|_{\dot{H}^1} = C\}, \quad (2.28)$$

where $C > 0$ is a constant. We construct the manifold by fixing the $\dot{H}^1$ seminorm because of the form of our objective functional used in the optimization search, which

will be described in Chapter 3. The question we must answer now is what value of the constraint parameter C we use. To determine this we look to our first initial condition, the 3D Taylor Green Vortex (TGV). The TGV in the domain $\Omega := \mathbb{T}_L^3$ has the form

$$\boldsymbol{\eta}_{TGV}(\mathbf{x}) = \begin{bmatrix} V_0 \sin(2\pi x/L) \cos(2\pi y/L) \cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L) \sin(2\pi y/L) \cos(2\pi z/L) \\ 0 \end{bmatrix}, \quad (2.29)$$

for some $V_0 \neq 0$. The $\dot{H}^1$ seminorm of (2.29) is

$$\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \sqrt{3}L\pi V_0, \quad (2.30)$$

which is proven in Appendix E. The BKM criterion (1.1.2) uses the quantity $\|\boldsymbol{\omega}(t)\|_{L^\infty}$ to determine singularity formation. For the TGV

$$\|\boldsymbol{\omega}\|_{L^\infty} = \|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = 4\pi V_0/L, \quad (2.31)$$

which is proven in Appendix F. In [5], the authors provided numerical evidence that blow-up of (1.2) may occur around $T^* \approx 4$ for initial condition (2.29) with parameters $L = 2\pi$ and $V_0 = 1$. This result provides the baseline for our study and so we must match this time window so that singularity formation identified by (2.1.3), if it happens, occurs at approximately the same time as in [5]. Therefore, we set $\lambda = 1$. Our Euler-Voigt system is defined on the domain Ω with $L = 1$ and so we set $\beta = 1/2\pi$. Therefore, from (2.27) we set $\gamma = 1/2\pi$. The model we are aiming to match uses $V_0 = 1$ and so to match the time window we use $V_0 = 1/2\pi$.

These considerations give $\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \sqrt{3}/2$ and $\|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = 2$ for the initial condition (2.29). Therefore, we can restrict our discussion of the TGV to initial conditions with $\dot{H}^1$ seminorm which belong to a closed manifold (2.28) with $C = \sqrt{3}/2$. In order to select initial conditions within this manifold, we will first select initial conditions $\boldsymbol{\eta}_\alpha \in V$ and then apply a retraction function to ensure it has $\|\boldsymbol{\eta}_\alpha\|_{\dot{H}^1} = \sqrt{3}/2$. To verify that this selection of the $\dot{H}^1$ seminorm of the initial condition is correct, we compare this scaling to the results shown in Figure 1b of [5]. This figure shows the evolution of the L^∞ norm of the vorticity over time.

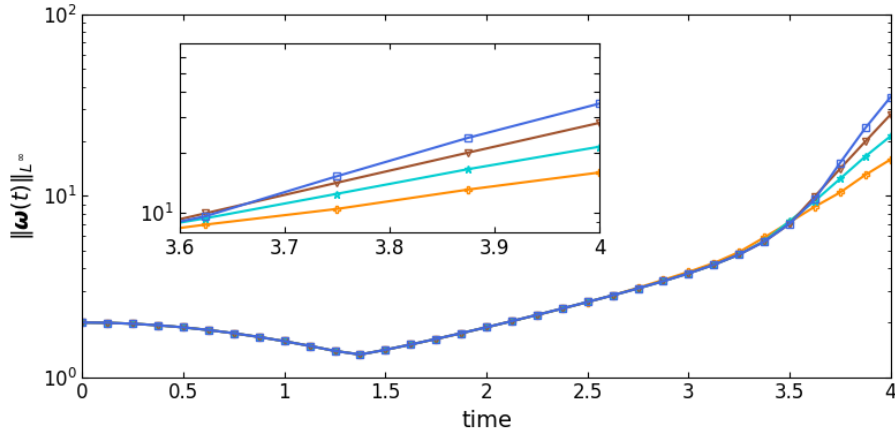


Figure 2.1: Maximum of vorticity $\|\boldsymbol{\omega}(t)\|_{L^\infty}$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), 1024^3 (blue squares).

We present our reproduction in Figure 2.1 which shows that the evolution of the vorticity at each resolution in our scaled model matches exactly to the corresponding computations shown in Figure 1b in [5]. Therefore, our construction of the manifold $\mathcal{M}_C$ is correct.

3 The Optimization Problem

In this chapter we **formulate the main** optimization problem and **introduce the** Riemannian conjugate gradient method used to solve it.

3.1 Formulation of the Problem

With our initial condition and blowup criterion defined by Theorem 2.1.3, we now construct the PDE-constrained optimization problem. We begin with the objective functional $\Phi_{\alpha,T} : V \rightarrow \mathbb{R}^+$, which is defined as

$$\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) := \|\nabla \mathbf{u}_\alpha(T; \boldsymbol{\eta}_\alpha)\|_{L^2}^2, \quad (3.32a)$$

$$= \|\mathbf{u}_\alpha(T; \boldsymbol{\eta}_\alpha)\|_{H^1}^2, \quad (3.32b)$$

where $\mathbf{u}_\alpha(T; \boldsymbol{\eta}_\alpha)$ is the solution of the Euler-Voigt equations (1.4) obtained at time T with the initial condition $\boldsymbol{\eta}_\alpha \in V$ and regularization parameter α . Using this objective functional, we define the optimization problem.

Problem 3.1.1. *Given $\alpha, T \in \mathbb{R}^+$, find*

$$\tilde{\boldsymbol{\eta}}_{\alpha,T} = \operatorname{argmax}_{\boldsymbol{\eta}_\alpha^0 \in M_C} \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0), \quad (3.33)$$

where $\boldsymbol{\eta}_\alpha^0$ is our initial guess. Our goal is to first fix T and solve the optimization problem for a sequence of decreasing values of α that converge to 0. From Figure 1b of [5], the expected time of singularity formation in the Euler flow, ~~with the scaled TGV as discussed in Section 2.3 used as the initial condition~~, is $T^* \approx 4$. From Theorem 2.1.3, if the initial condition criterion (2.8) and objective functional criteria (2.12) are both satisfied, then the corresponding Euler flow develops a singularity in the interval $[0, T^*]$ from initial condition $\boldsymbol{\eta}$. We examine the results for a time interval ~~slightly shorter and longer than T^*~~ . Therefore, we consider the optimization problem for $T = 3$, and $T = 5$. ~~With optimization, blow-up may occur in both cases.~~

3.2 Initial Guesses

Problem 3.1.1 is non-convex and so may ~~converge to a different local maximum~~ depending on the initial ~~condition~~ used at iteration zero, $\boldsymbol{\eta}_\alpha^0$. In order to search for optimal initial conditions that may ~~form~~ a singularity in the Euler flow, we must begin ~~the~~ optimization problem from different initial guesses. Before we begin our search for singularities, we examine the structure of the initial guesses used in this study. These include the TGV as well as two ~~optimized~~ initial conditions from the results of [30].

Figure 3.1 shows the isosurfaces of the magnitude of the **vorticity for $\boldsymbol{\eta}_{TGV}$** . The first initial condition from [30] is the **optimal initial condition found using the short time window and a resolution of 128^3 , using the TGV as the initial guess**. Figure 3.2 shows the isosurfaces of the magnitude of the vorticity for **$\boldsymbol{\eta}_S$** . These figures show that $\boldsymbol{\eta}_{TGV}$ is **symmetric along each axis** whereas $\boldsymbol{\eta}_S$ is concentrated in two vortex

tubes parallel to the x-axis.

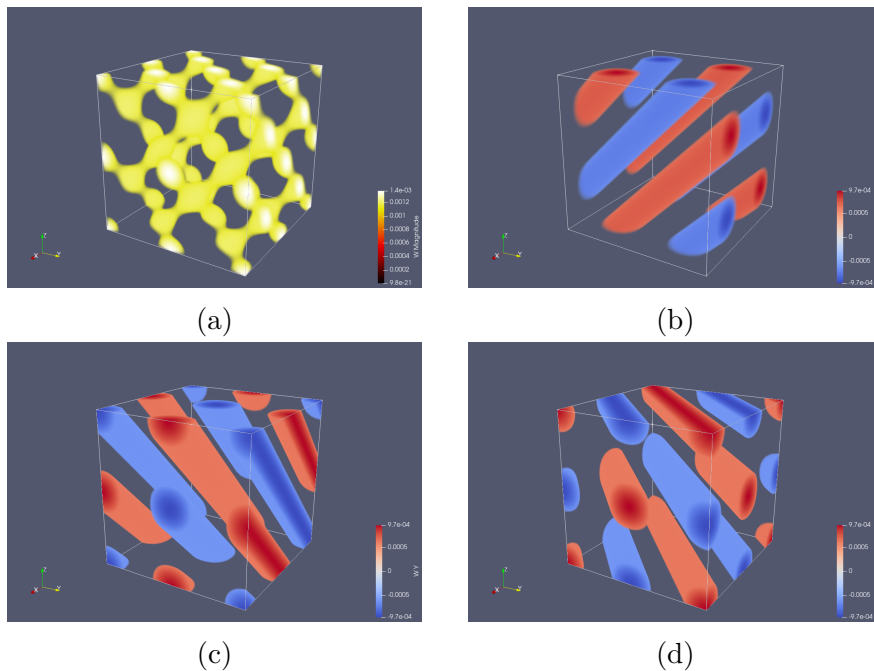


Figure 3.1: Isosurfaces of vorticity (a) magnitude $|\boldsymbol{\omega}(\boldsymbol{x})|$, and components (b) $|\omega_1|$, (c) $|\omega_2|$, and (d) $|\omega_3|$.

3.3 Gradient of Objective Function

The optimization problem (3.1.1) is solved using a Riemannian conjugate gradient method which uses an optimal search direction and distance to create the next approximation of the optimal initial condition. This method is utilized because it accelerates the gradient ascent method which is necessary given the computational cost of each iteration. To solve this problem, we utilize a 'optimize-then-discretize' approach where the optimization problem, and the Riemannian conjugate gradient method, are derived in an infinite-dimensional function space and then discretized to solve numerically. The same method was used in [30] to solve a PDE-constrained optimization

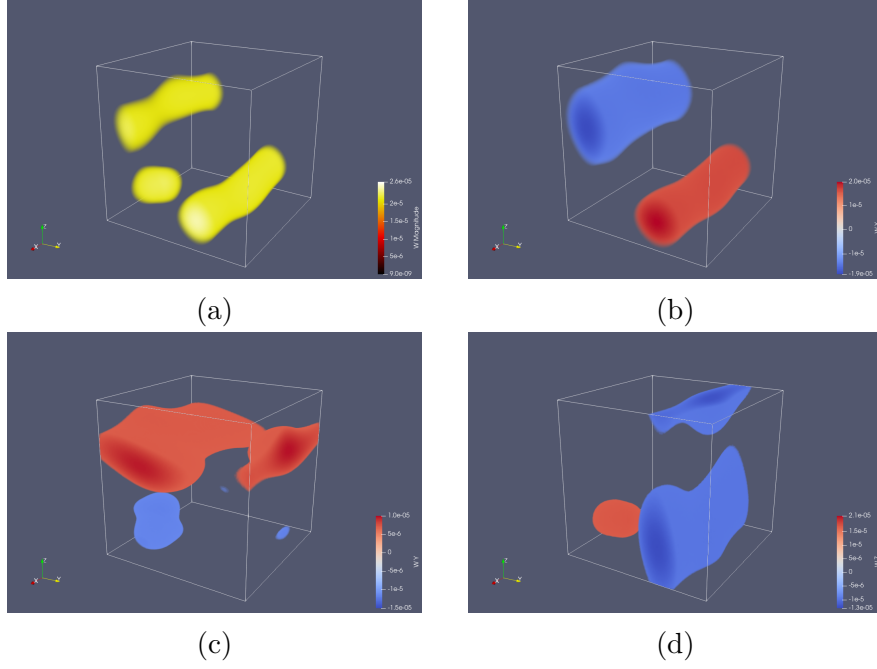


Figure 3.2: Isosurfaces of vorticity (a) magnitude $|\boldsymbol{\omega}(\mathbf{x})|$, and components (b) $|\omega_1|$, (c) $|\omega_2|$, and (d) $|\omega_3|$.

problem for the Euler equations to search for initial conditions $\boldsymbol{\eta}$ that produce 'extreme' flows.

We compute the optimal search direction using the gradient of the objection functional $\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)$. It is obtained using an adjoint method which depends on solving a properly defined adjoint system backwards in time. To derive an expression for the gradient, we first introduce the Gâteaux (directional) differential $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) : V \times V \rightarrow \mathbb{R}$. The Gateaux differential is defined as,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) := \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon \boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)]. \quad (3.34)$$

This represents the change in $\Phi_{\alpha,T}$ from an infinitesimal perturbation in the direction $\boldsymbol{\eta}'_{\alpha}$ from the initial condition $\boldsymbol{\eta}_{\alpha}$. By fixing the initial condition $\boldsymbol{\eta}_{\alpha}$, we can view the Gâteaux differential as a bounded linear functional on V . Therefore, it can also be represented as an inner product on the Gevrey space using the Riesz representation theorem [30, 23]

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_{\alpha}; \boldsymbol{\eta}'_{\alpha}) = \langle \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_{\alpha}), \boldsymbol{\eta}'_{\alpha} \rangle_{G^{\sigma}}. \quad (3.35)$$

Substituting the second definition of the objective functional (3.32b) into the **definition of the** Gâteaux differential (3.34), we derive the Gâteaux differential in the form of the $\dot{H}^1$ inner product,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_{\alpha}; \boldsymbol{\eta}'_{\alpha}) = 2 \langle \mathbf{u}_{\alpha}(T), \mathbf{u}'_{\alpha}(T) \rangle_{\dot{H}^1} \quad (3.36)$$

The derivation for this is shown in Appendix I. In (3.36) the value $\mathbf{u}'_{\alpha}$ is the solution of the linearization of the Euler-Voigt system (1.4) around its solution $\mathbf{u}_{\alpha}$ corresponding to the initial condition $\boldsymbol{\eta}_{\alpha}$. The linearization of the Euler-Voigt system is defined as

$$\mathcal{L} \begin{bmatrix} \mathbf{u}'_{\alpha} \\ p'_{\alpha} \end{bmatrix} := \begin{bmatrix} \partial_t \mathbf{u}'_{\alpha} + (I - \alpha^2 \Delta)^{-1} (\mathbf{u}_{\alpha} \cdot \nabla \mathbf{u}'_{\alpha} + \mathbf{u}'_{\alpha} \cdot \nabla \mathbf{u}_{\alpha} + \nabla p'_{\alpha}) \\ \nabla \cdot \mathbf{u}'_{\alpha} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (3.37a)$$

$$\mathbf{u}'_{\alpha}(x, 0) = \boldsymbol{\eta}'_{\alpha} \quad (3.37b)$$

where $\boldsymbol{\eta}'_{\alpha}$ and p'_{α} are the perturbations of the initial condition and pressure, respectively. The derivation of the linearized form of the Euler-Voigt equations is shown in Appendix G. In order to obtain the gradient of the objective functional from the Gâteaux differential we must first introduce the fields $\{\mathbf{u}_{\alpha}^*, p_{\alpha}^*\}$ which are the solution

of the adjoint system

$$\mathcal{L}^* \begin{bmatrix} \mathbf{u}_\alpha^* \\ p_\alpha^* \end{bmatrix} := \begin{bmatrix} -\partial_t \mathbf{u}_\alpha^* - (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}_\alpha^* + (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}_\alpha^*)^\top) \mathbf{u}_\alpha - \nabla p_\alpha^* \\ -\nabla \cdot \mathbf{u}_\alpha^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (3.38a)$$

$$\mathbf{u}_\alpha^*(\mathbf{x}, T) = 2|D|^2 \mathbf{u}_\alpha(\mathbf{x}, T) \quad (3.38b)$$

which is derived in Appendix H by performing integration by parts, in both space and time, on the following duality-pairing relation

$$\begin{aligned} \left(\mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix}, \begin{bmatrix} \mathbf{u}_\alpha^* \\ p'_\alpha \end{bmatrix} \right) &:= \int_0^T \int_{\mathbb{T}^3} \mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix}, \begin{bmatrix} \mathbf{u}_\alpha^* \\ p'_\alpha \end{bmatrix} d\mathbf{x} dt \\ &= \left(\begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix}, \mathcal{L}^* \begin{bmatrix} \mathbf{u}_\alpha^* \\ p'_\alpha \end{bmatrix} \right) + \int_{\mathbb{T}^3} \mathbf{u}'_\alpha(\mathbf{x}, T) \cdot \mathbf{u}_\alpha^*(\mathbf{x}, T) d\mathbf{x} \\ &\quad - \int_{\mathbb{T}^3} \mathbf{u}'_\alpha(\mathbf{x}, 0) \cdot \mathbf{u}_\alpha^*(\mathbf{x}, 0) d\mathbf{x} \\ &= 0. \end{aligned} \quad (3.39)$$

Using the solutions of the adjoint system we can now obtain a convenient expression for the gradient. **With** the Gâteaux differential (3.36) expressed in terms of the L^2 inner product

$$\Phi'_{\alpha, T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \langle \mathbf{u}_\alpha^*(0), \mathbf{u}'_\alpha(0) \rangle_{L^2}, \quad (3.40)$$

which is derived in Appendix I, and in the Riesz representation form (3.35), we obtain

$$\Phi'_{\alpha, T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \langle \mathbf{u}_\alpha^*(0), \mathbf{u}'_\alpha(0) \rangle_{L^2} = \langle \nabla \Phi_{\alpha, T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma}. \quad (3.41)$$

Using the definition of the Gevrey inner product (2.22), the L^2 inner product can be converted to the Gevrey inner product,

$$\langle \mathbf{u}_\alpha^*(0), \mathbf{u}'_\alpha(0) \rangle_{L^2} = \langle (1 + |D|^2)^{-3} e^{-2\sigma|D|} \mathbf{u}_\alpha^*(0), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma}. \quad (3.42)$$

Therefore, we have two forms of the Gâteaux differential as a Gevrey inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \langle \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma} = \langle (1 + |D|^2)^{-3} e^{-2\sigma|D|} \mathbf{u}_\alpha^*(0), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma}. \quad (3.43)$$

From this we obtain the gradient of the objective functional

$$\nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) = (1 + |D|^2)^{-3} e^{-2\sigma|D|} \mathbf{u}_\alpha^*(0). \quad (3.44)$$

The adjoint system (3.38) is linear and **represents** a terminal value problem. Its coefficients are determined by evolving the Euler-Voigt system forward from the initial condition $\boldsymbol{\eta}_\alpha$ to compute $\mathbf{u}_\alpha(T)$, from which we obtain the terminal value of (3.38). Finally, we evolve the adjoint system backwards in time to obtain the initial value $\mathbf{u}_\alpha^*(0)$ which is used to compute the gradient (3.44).

3.4 Riemannian Conjugate Gradient Method

When solving the optimization problem, we look for an initial condition $\boldsymbol{\eta}_\alpha \in M_C$ that maximizes our objective functional. Therefore, not only must our initial guess $\boldsymbol{\eta}_\alpha^0$ have a $\dot{H}^1$ seminorm of $\sqrt{3}/2$, but so too must each following **iterate** $\boldsymbol{\eta}_\alpha^{(n)}$. We ensure this constraint is satisfied using a Riemannian conjugate gradient method modified from the method used in [11, 28, 30]. This method has three steps:

1. Project the gradient $\nabla\Phi_T(\boldsymbol{\eta}_\alpha)$ onto the tangent space to $\mathcal{M}_C$ at $\boldsymbol{\eta}_\alpha$.
2. Perform a suitable vector transport operation in order to construct a Riemannian conjugate ascent direction using the previous search direction.
3. Retract the resulting state back to the constraint manifold $\mathcal{M}_c$.

We first define the space tangent to $\mathcal{M}_C$ at $\mathbf{v}$ as

$$T_v\mathcal{M}_C = \{\mathbf{z} \in V : \langle \mathbf{v}, \mathbf{z} \rangle_{\dot{H}^1} = 0\}. \quad (3.45)$$

For any $\mathbf{z} \in \mathcal{T}_v\mathcal{M}_C$, this condition can be expressed as

$$\langle \mathbf{v}, \mathbf{z} \rangle_{\dot{H}^1} = \int_{\mathbb{T}^3} |D| \mathbf{v} \cdot |D| \mathbf{z} d\mathbf{x} \quad (3.46a)$$

$$= \int_{\mathbb{T}^3} (1 + |D|^2)^3 e^{2\sigma|D|} \left((1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v} \right) \cdot \mathbf{z} d\mathbf{x} \quad (3.46b)$$

$$= \langle (1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}, \mathbf{z} \rangle_{G^\sigma} = 0. \quad (3.46c)$$

Thus, at each point $\mathbf{v} \in M_C$, the tangent space $\mathcal{T}_v\mathcal{M}_C$ can be characterized using the unit 'normal' vector given by

$$\mathbf{n}_v = \frac{(1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}}{\|(1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}\|_{G^\sigma}} \quad (3.47)$$

which allows us to construct the projection operator $\mathcal{P}_{\mathcal{T}_v\mathcal{M}_C} : V \rightarrow \mathcal{T}_v\mathcal{M}_C$,

$$\forall \boldsymbol{\omega} \in V, \quad \mathcal{P}_{\mathcal{T}_v\mathcal{M}_C} \boldsymbol{\omega} = \boldsymbol{\omega} - \langle \mathbf{n}_v, \boldsymbol{\omega} \rangle_{G^\sigma} \mathbf{n}_v. \quad (3.48)$$

As stated previously, the initial conditions at each iteration must have a $\dot{H}^1$ seminorm of $\sqrt{3}/2$. In order to ensure this we introduce the retraction operator

$$R(\mathbf{v}) = \frac{\sqrt{3}}{2} \frac{\mathbf{v}}{\|\mathbf{v}\|_{\dot{H}^1}}, \quad \mathbf{v} \neq \mathbf{0}. \quad (3.49)$$

The conjugate-gradient method method uses the search direction obtained at the n -th iteration to compute the search direction at the $(n+1)$ -th iteration. This is done by computing a linear combination of the gradient at the current iteration and the search direction from the previous iteration, weighted with a 'momentum' coefficient. However, the two directions belong to different tangent spaces to $\mathcal{M}_C$ and thus cannot be directly added. In order to allow algebraic operations to be performed on the vectors of the two tangent spaces we introduce the vector transport operation defined by [1]

$$\mathcal{TM}_C \oplus \mathcal{TM}_C \rightarrow \mathcal{TM}_C \quad : \quad (\boldsymbol{\varphi}, \boldsymbol{\xi}) \mapsto \Gamma_{\boldsymbol{\varphi}}(\boldsymbol{\xi}) \in \mathcal{TM}_C,$$

where $\mathcal{TM}_C = \bigcup_{\mathbf{v} \in \mathcal{M}_C} \mathcal{T}_{\mathbf{v}}\mathcal{M}_C$ is the tangent bundle on $\mathcal{M}_C$. The operation transports the vector field $\boldsymbol{\xi}$ along the manifold $\mathcal{M}_C$ in the direction of $\boldsymbol{\varphi}$. This allows us to map search directions between the tangent spaces of two consecutive iterations. This operator is defined via differentiated retraction

$$\Gamma_{\boldsymbol{\varphi}_{\mathbf{u}}}(\boldsymbol{\xi}_{\mathbf{u}}) := \frac{d}{dt} R(\boldsymbol{\varphi}_{\mathbf{u}} + t\boldsymbol{\xi}_{\mathbf{u}}) \Big|_{t=0} \quad (3.50a)$$

$$:= \frac{\sqrt{3}}{2} \frac{1}{\|\mathbf{u} + \boldsymbol{\varphi}_{\mathbf{u}}\|_{\dot{H}^1}} \left[\boldsymbol{\xi}_{\mathbf{u}} - \frac{\langle \mathbf{u} + \boldsymbol{\varphi}_{\mathbf{u}}, \boldsymbol{\xi}_{\mathbf{u}} \rangle_{\dot{H}^1}}{\|\mathbf{u} + \boldsymbol{\varphi}_{\mathbf{u}}\|_{\dot{H}^1}^2} (\mathbf{u} + \boldsymbol{\varphi}_{\mathbf{u}}) \right], \quad \mathbf{u} \in \mathcal{M}_C. \quad (3.50b)$$

With our operators defined we can now construct the iterative relation that will be used to maximize our objective functional (3.32)

$$\boldsymbol{\eta}_\alpha^{(n+1)} = R \left[\boldsymbol{\eta}_\alpha^{(n)} + \tau_n \mathbf{d}^{(n)} \right]. \quad (3.51)$$

To simplify notation, let $\mathbf{G}^{(n)} := \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^{(n)})$ and let $\mathcal{T}_n := \mathcal{T}_{\boldsymbol{\eta}_\alpha^{(n)}} \mathcal{M}_C$. The search direction at iteration n ($\mathbf{d}^{(n)}$) is computed as

$$\begin{aligned} \mathbf{d}^{(0)} &= P_{\mathcal{T}_0} \mathbf{G}^{(0)} \\ \mathbf{d}^{(n)} &= P_{\mathcal{T}_n} \mathbf{G}^{(n)} + \beta_n \Gamma_{\tau_{n-1}} \mathbf{d}^{(n-1)}(\mathbf{d}^{(n-1)}), \quad n \geq 1, \end{aligned}$$

where the 'momentum' term is computed using the Polak-Ribière approach [27].

$$\beta_n = \frac{\left\langle P_{\mathcal{T}_n} \mathbf{G}^{(n)}, \left(P_{\mathcal{T}_n} \mathbf{G}^{(n)} - \Gamma_{\tau_{n-1}} \mathbf{d}^{(n-1)} P_{\mathcal{T}_{n-1}} \mathbf{G}^{(n-1)} \right) \right\rangle_{G^\sigma}}{\|P_{\mathcal{T}_{n-1}} \mathbf{G}^{(n-1)}\|_{G^\sigma}^2}. \quad (3.53)$$

In order to remove old data that may slow down convergence, the algorithm is periodically restarted by setting $\beta_n = 0$. For our optimization search, we restart the algorithm every 20 iterations, and when the search direction is not an ascent direction, i.e, when

$$\frac{\left\langle \mathbf{d}^{(n)}, P_{\mathcal{T}_n} \mathbf{G}^{(n)} \right\rangle_{G^\sigma}}{\|\mathbf{d}^{(n)}\|_{G^\sigma} \|P_{\mathcal{T}_n} \mathbf{G}^{(n)}\|_{G^\sigma}} < 0. \quad (3.54)$$

Finally, our optimal step size is determined by the distance along **an arc tangent to $\mathbf{d}^{(n)}$** which maximizes the objective functional, it is computed by performing an arc-search along this direction using a modification of Brent's method

$$\tau_n = \operatorname{argmax}_{\tau > 0} \Phi_T \left(R \left[\boldsymbol{\eta}_\alpha^{(n)} + \tau \mathbf{d}^{(n)} \right] \right). \quad (3.55)$$

The optimization search is terminated when the relative change in the objective functional from (3.51) is below a chosen tolerance.

$$0 \leq \frac{\Phi_{\alpha,T}(\boldsymbol{\eta}_{\alpha}^{(n+1)}) - \Phi_{\alpha,T}(\boldsymbol{\eta}_{\alpha}^{(n)})}{\Phi_{\alpha,T}(\boldsymbol{\eta}_{\alpha}^{(n)})} < tol, \quad (3.56)$$

where $tol = 10^{-10}$.



4 Numerical Methods

In this chapter we describe the numerical techniques used to solve the optimization problem. In addition, we determine the parameters including the resolution, time step size, regularization parameter α , and Gevrey parameters s and σ .

4.1 Discretization & Spectral Methods

~~Simulations were computed~~ using a standard pseudospectral fourier method on ~~a periodic unit cube, discretized in~~ a uniform grid of spatial resolutions $N = 128^3, 256^3, 512^3, 1024^3$. Derivatives are evaluated in the Fourier space and nonlinear products are computed in the physical space. The standard 2/3 dealiasing rule is also applied.

In [20], ~~the authors use a range of~~ $\alpha = 12/1024, \dots, 36/1024$. We extend this range in both directions and use the regularization parameters $\alpha = 8/1024, \dots, 64/1024$.

Time stepping is performed with the explicit fourth-order Runge-Kutta (RK4) method. The determination of our time step size Δt is based on two factors. First, we consider the accuracy of the solution, determined by the relative error of the objective functional (3.32) to the results for a minimum step size for a time interval $T = 5$.

Second, we consider the conservation of the α -energy (1.6), determined by the maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$, for $t \in [0, 5]$. These factors are analysed for each resolution and value of α for the TGV.

The goal of our optimization search is to identify initial conditions $\tilde{\eta}_{\alpha,T}$ that produce 'extreme' behaviour in solutions to the Euler-Voigt equations. Therefore, the TGV will by definition be less extreme than the initial conditions at each iteration of the optimization search. To ensure stability, accuracy, and conservation of α -energy are maintained upon optimization we must select a step size smaller than the minimum step size identified by the above criteria.

4.1.1 Numerical Stability & Accuracy

We begin with the accuracy of the Euler-Voigt model. To determine the accuracy of the model at each time step size we start from the TGV and evaluate the objective functional (3.32) for a time interval of $T = 5$. First, we do this for $\Delta t = 2^{-8}$ and designate this result as our 'correct' value Φ_c . We then compute the relative error of the objective functional for each value of Δt ($\Phi_{\Delta t}$) to Φ_c

$$\epsilon_\Phi = \left| \frac{\Phi_c - \Phi_{\Delta t}}{\Phi_c} \right|. \quad (4.57)$$

Table (4.1) shows the results of ϵ_Φ for $\alpha = 16/1024, \dots, 64/1024$. We include only these values of α in this table because they had identical results for all four resolutions. This is due to the fact that the regularization term of (1.4) kept the solution well-resolved even at the lowest resolution, and so the accuracy did not increase with the higher resolutions. Table (4.2) shows the results of ϵ_Φ for $\alpha = 8/1024, 12/1024$. For these

		α				
		16/1024	24/1024	32/1024	48/1024	64/1024
Δt	2^{-2}	7.00E-05	1.45E-05	4.80E-06	8.70E-07	1.58E-07
	2^{-3}	3.64E-06	7.89E-07	2.63E-07	5.03E-08	9.29E-09
	2^{-4}	2.02E-07	4.53E-08	1.52E-08	3.01E-09	5.63E-10
	2^{-5}	1.21E-08	2.70E-09	9.16E-10	1.83E-10	3.48E-11
	2^{-6}	8.63E-10	1.65E-10	5.99E-11	8.33E-12	2.05E-12
	2^{-7}	9.44E-11	9.42E-12	5.99E-12	0	0

Table 4.1: ~~Results~~ ϵ_Φ for $\alpha = 16/1024, \dots, 64/1024$.

values the model becomes under-resolved at the lowest resolution and so accuracy does increase at the higher resolutions.

		α	
		8/1024	12/1024
Δt	2^{-2}	0.0045	2.66E-04
	2^{-3}	0.0011	1.46E-05
	2^{-4}	4.85E-04	1.30E-06
	2^{-5}	2.33E-04	3.19E-07
	2^{-6}	1.02E-04	1.23E-07
	2^{-7}	3.45E-05	4.05E-08

(a) $N = 128^3$

		α	
		8/1024	12/1024
Δt	2^{-2}	0.003	2.64E-04
	2^{-3}	1.82E-04	1.34E-05
	2^{-4}	9.46E-06	7.05E-07
	2^{-5}	5.12E-07	3.95E-08
	2^{-6}	3.00E-08	2.32E-09
	2^{-7}	1.96E-09	1.31E-10

(b) $N = 256^3, 512^3, 1024^3$

Table 4.2: ~~Results~~ ϵ_Φ for $\alpha = 8/1024, 12/1024$.

We now examine the conservation of α -energy for the Euler-Voigt model. To determine the accuracy of the model at each time step size we start from the TGV and

		α				
		16/1024	24/1024	32/1024	48/1024	64/1024
Δt	2^{-2}	8.73E-06	1.90E-06	5.95E-07	9.75E-08	1.64E-08
	2^{-3}	3.72E-07	7.68E-08	2.29E-08	3.81E-09	6.05E-10
	2^{-4}	1.76E-08	3.47E-09	9.74E-10	1.67E-10	2.45E-11
	2^{-5}	9.59E-10	1.74E-10	4.71E-11	9.79E-12	2.80E-12
	2^{-6}	7.55E-11	9.29E-12	3.27E-12	2.41E-12	1.96E-12
	2^{-7}	1.47E-11	2.45E-12	2.86E-12	3.12E-12	2.01E-12
	2^{-8}	6.00E-12	3.07E-12	3.12E-12	3.55E-12	2.21E-12

Table 4.3: Results ϵ_α for $\alpha = 16/1024, \dots, 64/1024$ with $N = 128^3, 256^3, 512^3, 1024^3$.

evaluate the maximum ~~of the~~ relative error of the α -energy:

$$\epsilon_\alpha = \max_{t \in [0, T]} \left| \frac{E_\alpha(t) - E_\alpha(0)}{E_\alpha(0)} \right|. \quad (4.58)$$

Table (4.3) shows the ~~results~~ of ϵ_α for $\alpha = 16/1024, \dots, 64/1024$. Just as with the results of ϵ_Φ , we include only these values of α because they ~~had~~ identical results for all four resolutions. Table (4.4) shows the ~~results~~ of ϵ_α for $\alpha = 8/1024, 12/1024$. The time step size of each model run was selected based on the results ~~of~~ these tables to ensure accuracy and conservation of the α -energy for the chosen α and resolution.

4.2 ~~κ Test Validation~~

In the previous sections we examined the accuracy of the objective functional (3.32) and α -energy (1.6) based on the resolution N , regularization parameter α , and time step size Δt . ~~The κ test is used to evaluate the accuracy of the gradient calculation,~~

		α	
		8/1024	12/1024
Δt	2^{-2}	3.82E-04	3.35E-05
	2^{-3}	6.91E-05	1.60E-06
	2^{-4}	3.05E-05	1.26E-07
	2^{-5}	1.57E-05	3.14E-08
	2^{-6}	8.05E-06	1.41E-08
	2^{-7}	4.08E-06	6.99E-09
	2^{-8}	2.05E-06	3.50E-09

(a) $N = 128^3$

		α	
		8/1024	12/1024
Δt	2^{-2}	2.91E-04	3.33E-05
	2^{-3}	1.49E-05	1.50E-06
	2^{-4}	6.85E-07	7.21E-08
	2^{-5}	3.40E-08	3.80E-09
	2^{-6}	1.89E-09	2.14E-10
	2^{-7}	1.38E-10	1.20E-11
	2^{-8}	2.37E-11	2.80E-12

(b) $N = 256^3, 512^3, 1024^3$

Table 4.4: ~~Results~~ ϵ_α for $\alpha = 8/1024, 12/1024$.

~~which is based on the validation method of [2, 30]. The value κ is defined as~~

$$\kappa(\epsilon) = \frac{\epsilon^{-1} [\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon \boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)]}{\langle \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma}}, \quad (4.59)$$

where the numerator is the Gâteaux differential (3.34), and the denominator is the Reisz form of the differential (3.35).

A log-log plot of $|\kappa - 1|$ vs ϵ can be broken up into three regions, which are dominated by roundoff errors, truncation errors, or discretization errors. The roundoff errors are caused by ~~finite precision numerical representation~~ and so are approximately of order $\mathcal{O}(\epsilon^{-1})$. Therefore, (4.59) will be dominated by round off errors for small ϵ values. The truncation errors are caused by the finite difference formula to ~~calculate~~ the numerator, and are of order $\mathcal{O}(\epsilon)$. Therefore, (4.59) will be dominated by the truncation errors for large values of ϵ . Between these two regions (4.59) will be ~~determined~~ by the

discretization errors, which are a product of the spatial resolution N and time step size Δt . Increasing the spatial resolution or decrease the time step size will reduce these errors. Therefore, if the calculation of the gradient is accurate, the log-log plot of $|\kappa - 1|$ vs ϵ will be of the general form shown in 4.1.

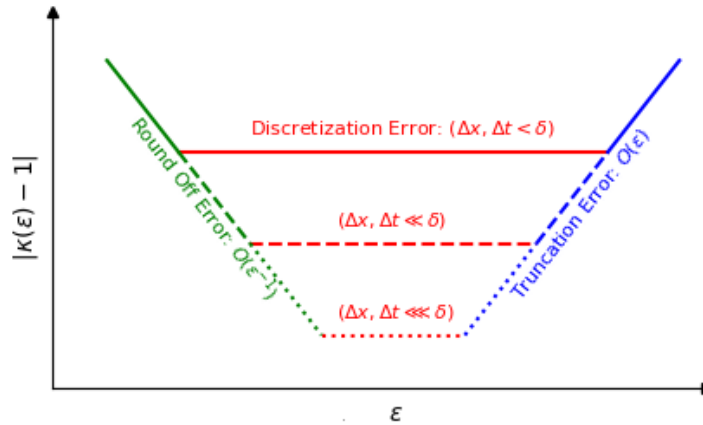
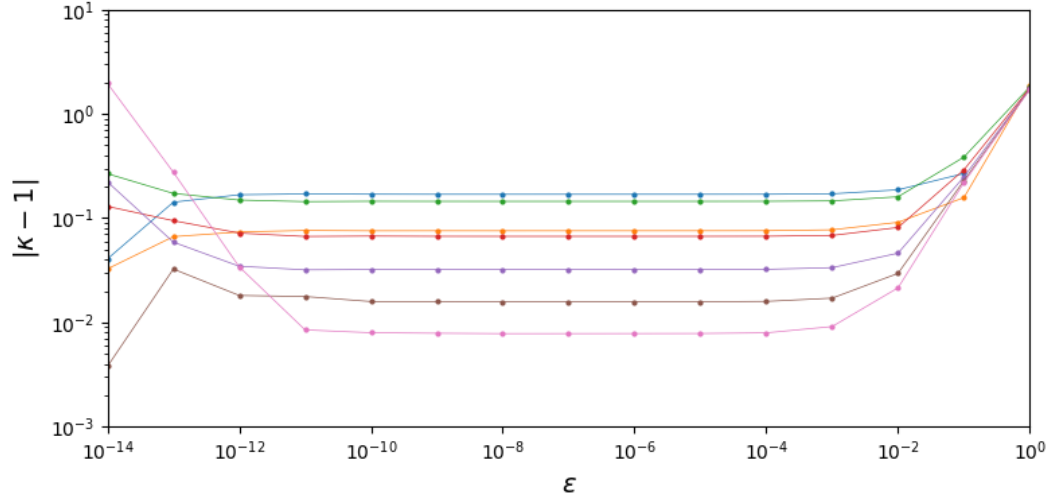
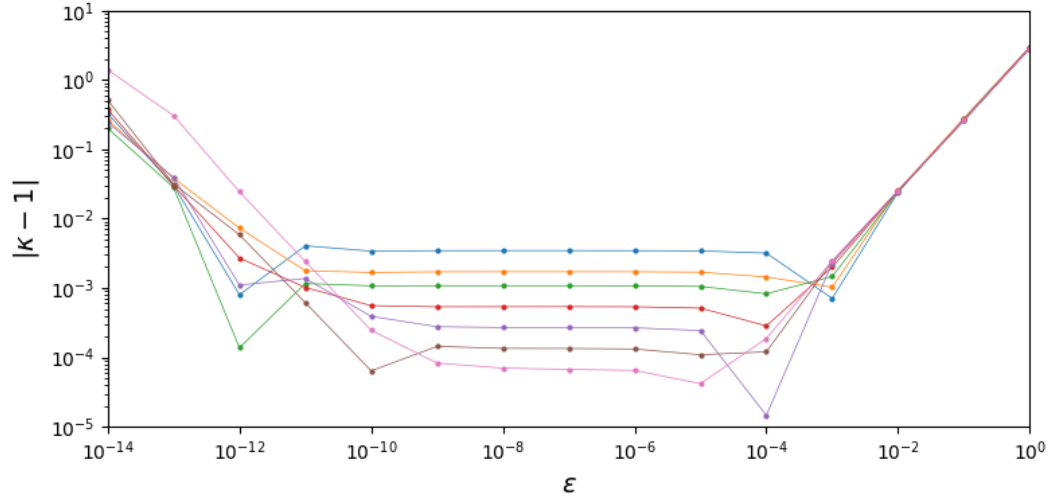


Figure 4.1: Theoretical κ behaviour.

The κ test was performed by fixing the spacial resolution, to keep the model well resolved, and varying the time step size $\Delta t \in [2^{-2}, 2^{-8}]$. The simulations were run for time window of $T = 5$ and $\epsilon \in [1\text{E-}14, 1]$. The results for $\alpha = 8/1024$ and $\alpha = 64/1024$ are shown in Figures 4.2 and 4.3, respectively.

The results from the κ test show the expected behaviour from the theoretical kappa test graph. Therefore, we conclude that the numerical techniques used to compute the gradient are accurate.

Figure 4.2: ~~κ test results~~ for $\alpha = 8/1024$.Figure 4.3: ~~κ test results~~ for $\alpha = 64/1024$.

4.3 Determination of ~~Gevrey class size~~

In Section 2.2 we introduced the Gevrey space (2.21) in which we search for optimal initial conditions. The ~~size~~ of the space is determined by two parameters, $s \geq 3$ and

$\sigma > 0$. We use a fixed value of $s = 3$. In order to determine the value of σ , we repeat the optimization problem for $\sigma = 1E-1, \dots, 1E-5$ and compare the objective functional of $\tilde{\eta}_{\alpha,T}$. The results for $\alpha = 16/1024$ and $T = 5$ are shown in Figure 4.4.

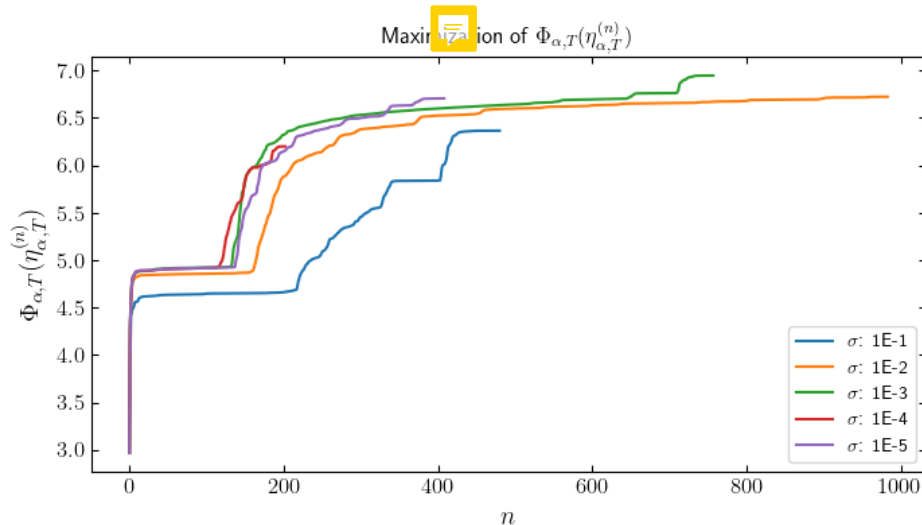


Figure 4.4: $\Phi_{\alpha,T}(\eta_{\alpha,T}^{(n)})$ vs iteration number n .

The results of Figure 4.4 show that the number of iterations it took to reach the maximum value does change depending on the value of σ , but there was not a significant change in the value of the objective functional. This was also the result of the other sigma tests for different values of α . These results are consistent with those of [30] which found that the optimization results did not significantly change so long as $\sigma < 1E-1$. Therefore, we make the same selection as in [30] and choose $\sigma = 1E-5$.

4.4 ~~Spectrum Validation~~

The energy spectrum of the solution of (1.4) is defined as,

$$e(k, t) := \frac{1}{2} \sum_{j \leq |j| < j+1} |\hat{\mathbf{u}}_j(t)|^2, \quad k = 2\pi j, \quad j \in \mathbb{N}. \quad (4.60)$$

The regularization term of (1.4) acts as a filter which suppresses the growth of small-scale structures of length-scale smaller than the regularization parameter α [26]. Due to this filter, the solution to (1.4) can remain well-resolved with relatively low resolution models. This is ideal for our analysis because it ensures that the model can fully capture the small scale structures of the fluid and so we can be certain that our objective functional (3.32) is accurate.

Figures 4.5 and 4.6 show the energy spectrum of the solution corresponding to the TGV over a time interval $T = 5$ at a resolution of $N = 128^3$ for $\alpha = 8/1024$ and $\alpha = 64/1024$, respectively. These figures demonstrate the exponential decay in the Fourier coefficients expected for an analytic function. As the solutions evolve in time the rate of decay decreases as energy is transferred to small scale structures. In Figure 4.5 a portion of the spectrum is cut-off by the 2/3 dealiasing rule. Due to the filter the solution for Figure 4.6 remains well resolved at only a resolution of $N = 128^3$.

In order to minimize the computational cost, each optimization search was started from the initial guess at the lowest resolution and ran until condition (3.56) was satisfied. If the optimal initial condition $\boldsymbol{\eta}_\alpha^{(n)}$ was under-resolved upon evolving over the

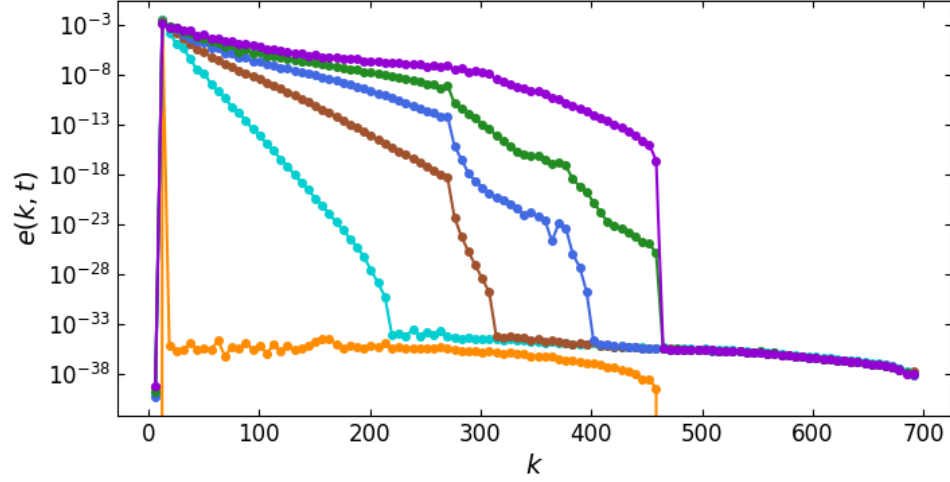


Figure 4.5: $e(k, t)$ vs. k for $\alpha = 8/1024$ at $t = 0$ (orange), $t = 1$ (cyan), $t = 2$ (brown), $t = 3$ (blue), $t = 4$ (green), $t = 5$ (purple).

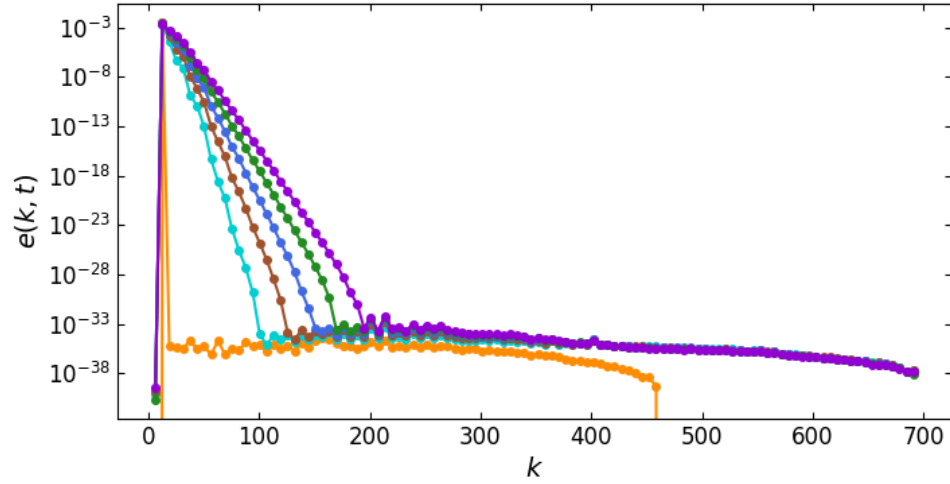


Figure 4.6: $e(k, t)$ vs. k for $\alpha = 64/1024$ at $t = 0$ (orange), $t = 1$ (cyan), $t = 2$ (brown), $t = 3$ (blue), $t = 4$ (green), $t = 5$ (purple).

time interval T , then the initial condition was up-scaled to the next higher resolution by padding with zeros in Fourier space and then converting back to physical space. Then the optimization search was started again, now with the up-scaled $\eta_\alpha^{(n)}$ as the

initial guess. Optimization was stopped when criterion (3.56) was satisfied and the initial condition $\tilde{\eta}_{\alpha,T}$ identified by solving Problem (3.1.1) remained well resolved when evolved over the time interval T . Using this method we can ensure that the initial conditions were all well-resolved, and so we can be certain our measurement of the objective functional is accurate for all of the optimal initial conditions discussed in Chapter 5.

5 Results

Utilizing the methods described in Chapters 2, 3, and 4, we present numerical evidence that a singularity does not form in the Euler flow **starting** from the Taylor Green Vortex initial condition. In addition, we present numerical evidence that upon optimization ... RESULTS!

5.1 Singularity Detection

Theorem 2.1.3 provides the criteria we use to determine if a singularity forms in the Euler flow corresponding to initial condition $\boldsymbol{\eta}$, where $\boldsymbol{\eta}$ is the optimal initial condition found in the limit $\alpha \rightarrow 0$. In practice, we take it as the optimal initial condition $\tilde{\boldsymbol{\eta}}_{\alpha,T}$ found for the smallest value of α . First, we check if the sequence of optimal initial conditions, found by solving Problem 3.1.1 for decreasing α , converge to a well-defined limit, i.e, if they satisfy (2.8). If this ~~criterion~~ is satisfied, we then check whether (2.12) is satisfied and $\alpha^2 \Phi_{\alpha,T}$ converges to a positive ~~value~~ as $\alpha \rightarrow 0$.

Simply plotting the **LHS** in (2.8) and (2.12) as functions of α will not be sufficient for determining if each condition is satisfied. This is because it is ~~not necessarily clear whether or not~~ these expressions are converging to zero or to some small positive

value. In order to make this determination we use the method of [20], which proposes the following ansatz,

$$\sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \approx \mathcal{O}(\alpha^p), \quad (5.61)$$

for $T^* > 0$ sufficiently large and for some power p . From this and Theorem 2.1.3, in order for a singularity to form in the Euler flow, it must be the case that the increase in the objective functional ~~must offset the decrease of α , which can only occur if $p \leq -1$. However, if $p < -1$, then the enstrophy term of 1.6 will become unbounded as $\alpha \rightarrow 0$. This would mean the α -energy is not conserved. Therefore, for a singularity to form, it must be the case that $p = -1$.~~ To find the value of p , we create a log-log plot of $\max_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}$ vs α . In order to simplify the visualization of the evolution of this quantity over time, the figures included here will only include the results at $T = 0, 1, \dots, T^*$. The value of p is defined as the steepest slope between any two values of α at any point in the time window.

We utilize the same method to determine if (2.8) is satisfied. However, in this case the condition is satisfied if the quantity does converge to zero. The condition is a sum of a energy term, $\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2$ and a enstrophy term, $\|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2$, which is scaled by α^2 . From Poincare's inequality, if the enstrophy term diminishes to zero as $\alpha \rightarrow 0$, then so too does the energy term and so (2.8) is satisfied [13]. However, if the enstrophy term diverges, but does so more slowly than α diminishes, then (2.8) is only satisfied if the energy term also goes to zero with α . Finally, if the enstrophy term diverges more quickly than α diminishes then (2.8) is not satisfied. Therefore,

we propose the following two ansätze.

$$\|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2} \approx \mathcal{O}(\alpha^q). \quad (5.62)$$

and

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2} \approx \mathcal{O}(\alpha^r), \quad (5.63)$$

By creating log-log plots of the LHS in (5.62) and (5.63) vs α , the values of q and r are given by the steepest slope between any two values of α at any point in time for (5.62) and (5.63), respectively. For (2.8) to be satisfied, it must be the case that either $w > 0$, or $-1 < w \leq 0$ and $q > 0$.

5.2 Results Obtained Using the Initial Guesses

We begin by presenting our results obtained using the initial guess, the TGV, without optimization, which serve as a baseline for optimization. A visualization of the vorticity of the TGV is shown in Figure 3.1. Figure 5.1 shows the vorticity $\|bn \times \mathbf{u}_\alpha(\mathbf{x}, T)$ of the solution at $T = 3$ for $\alpha = 8/1024$ and Figure 5.2 shows the vorticity of the solution at $T = 5$. The figures show that the areas of high vorticity are concentrated into smaller area as the flow evolves. They also show that the peak of the vorticity increases in magnitude.

Without optimization, the initial conditions for all values of α are simply the TGV. In this case, it is trivially true that condition (2.8) is satisfied, and so we only need to compute the value of p to determine if a singularity forms in the Euler flow corresponding to the TGV.

Figures 5.3 and 5.4 show that without optimization the steepest slope achieved

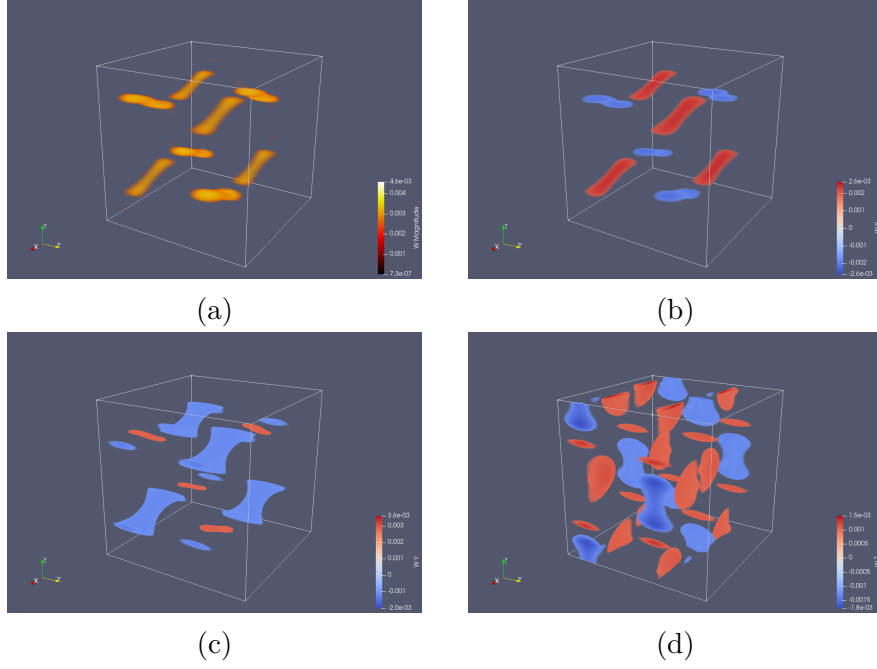


Figure 5.1: Isosurfaces of vorticity at $T = 3$. (a) magnitude $|\boldsymbol{\omega}(\mathbf{x})|$, and components (b) $|\omega_1|$, (c) $|\omega_2|$, and (d) $|\omega_3|$.

at any point in the time interval is $p = -0.225$ and this slope occurs at $T = 3$. Therefore, these results indicate that a singularity does not form in Euler within the time interval $[0, 3]$ from the TGV initial condition. They also show that the slope is still decreasing at the end of the time window, indicating that they may reach a slope of $p < -1$ at some later point in time. This result is consistent with the results of [5, 20] as the time interval is too short.

Figures 5.5 and 5.6 show that without optimization the steepest slope achieved at any point in the time is $p = -0.417$ and that this slope occurs at $T = 5$. This slope is larger than the p value for the shorter time window. However, it is still not close to the $p = -1$ required by (5.61). Therefore, these results indicate that a singularity does not form in Euler within the time interval $[0, 5]$ from the TGV initial condition.

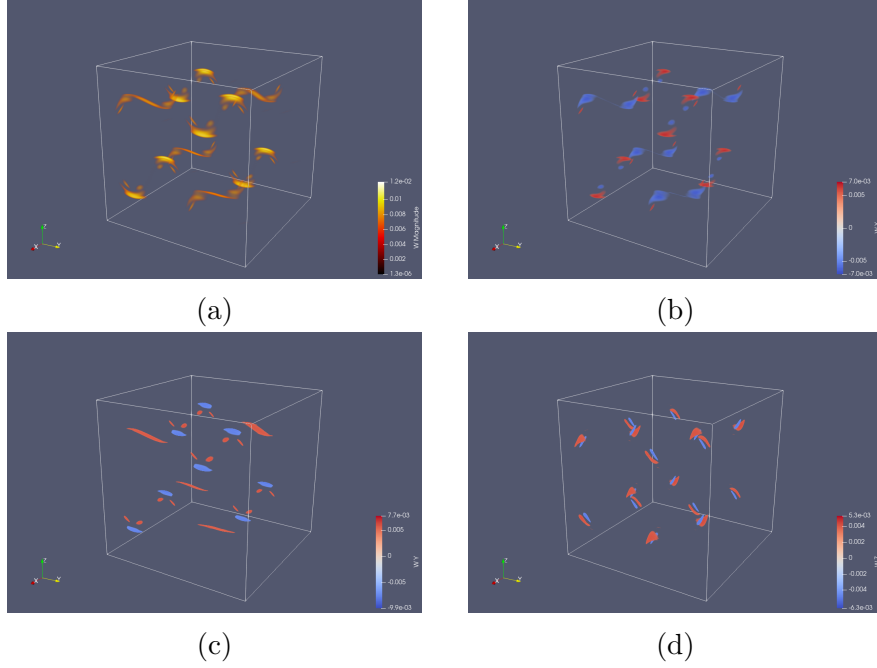


Figure 5.2: Isosurfaces of vorticity at $T = 5$. (a) magnitude $|\boldsymbol{\omega}(\mathbf{x})|$, and components (b) $|\omega_1|$, (c) $|\omega_2|$, and (d) $|\omega_3|$.

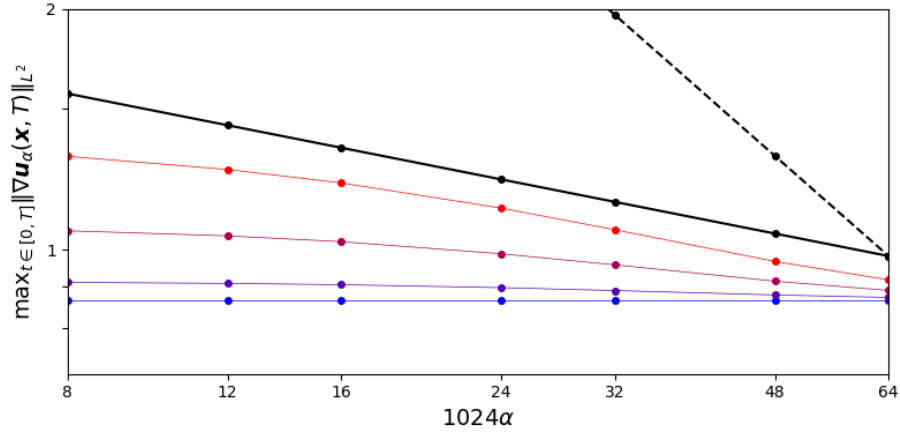


Figure 5.3: Solution of the Euler-Voigt system at $T = 3$ with the TGV used as the initial condition. Log-log plot of $\max_{t \in [0, T]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}$ vs α at $t = 0$ (blue) to $t = 3$ (red), Steepest slope $p = -0.225$ (black), slope of $p = -1$ (dotted black)

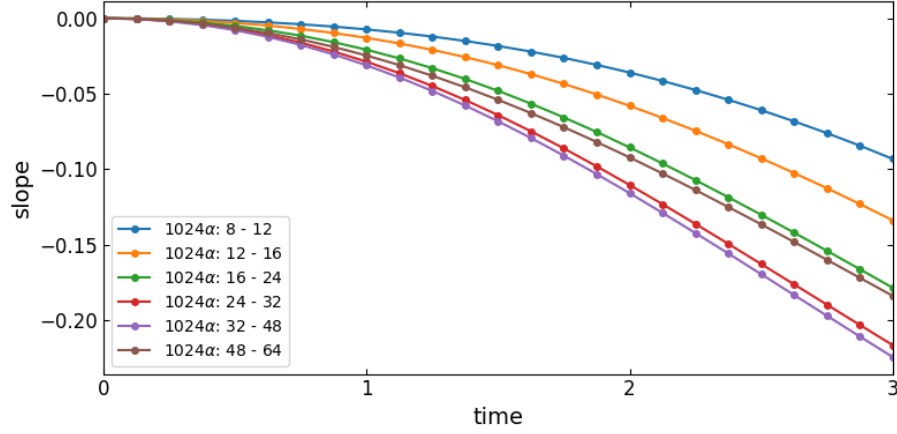


Figure 5.4: TGV for $T = 3$. Plot of the slope of $(\max_{t \in [0, T]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \text{ vs } \alpha)$ vs time for each α .

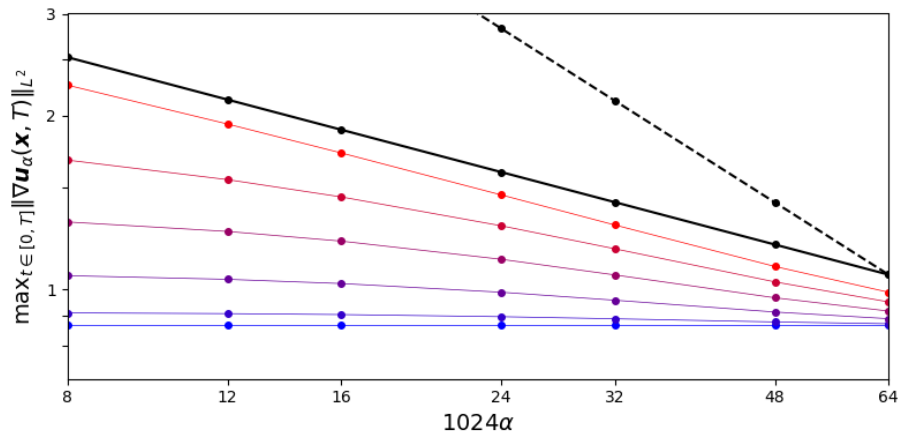


Figure 5.5: TGV for $T = 5$. Log-log plot of $\max_{t \in [0, T]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \text{ vs } \alpha$ at $t = 0$ (blue) to $t = 5$ (red), Steepest slope $p = -0.417$ (black), slope of $p = -1$ (dotted black)

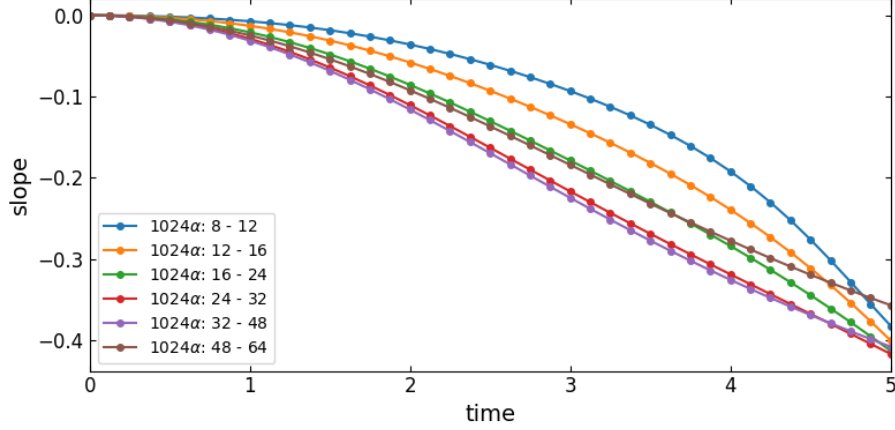


Figure 5.6: TGV for $T = 5$. Plot of the slope of $(\max_{t \in [0, T']} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \text{ vs } \alpha)$ vs time at each α .

This result is conflicts with the results of [5, 20], which indicate a singularity should form at $T^* \approx 4$.

5.2.1 Comparison to Larios et al. (2018)

Our results for the formation of a singularity in the Euler flow corresponding to the TGV initial condition conflict with those of [20]. In [20] their analysis also examined the Euler-Voigt equations in order to search for singularity formation in Euler from the TGV. As shown in Figure 3 of [20], they obtained a result of $p \approx 1$, which indicates that blow-up does occur in Euler flows starting from the TGV. Additionally, the minimum of p occurred at a time consistent with the results of [5]. However, their model used a 3D unit torus ($L = 1$) and a coefficient of $V_0 = 1$ in (2.29), rather than the $V_0 = 1/2\pi$ we use to match the scaling of [5]. Because of this the time scale of their result does not match the time scale of [5]. Therefore, the results of [20] do indicate a singularity forms in Euler from the TGV initial condition. However, the

blow-up time is much longer than the $T^* \approx 4$ of [5] which is consistent with our result.

6 Conclusion

Every thesis also needs a concluding chapter

A Function Spaces

In this Appendix we introduce the Lebesgue space, the space of continuous functions, and Sobolev space. These function spaces are used to define the initial conditions and solutions of the Euler and Euler-Voigt equations, the criteria for singularity formation, the objective functional, the gradient of the objective functional, and the Riemannian conjugate gradient method.

The Lebesgue space $L^p(\Omega)$ consists of real-valued measurable functions $\mathbf{u}$, defined on Ω , and $p \in \mathbb{R}^+$ for which the norm is bounded. The L^p norm is defined as

$$\|\mathbf{u}\|_{L^p} = \left(\int_{\Omega} |\mathbf{u}|^p d\mathbf{x} \right)^{1/p},$$

where $1 \leq p < \infty$. For $p = \infty$, the norm is given by

$$\|\mathbf{u}\|_{L^\infty} = \operatorname{ess\,sup}_{\mathbf{x} \in \Omega} |\mathbf{u}(\mathbf{x})|,$$

where $\operatorname{ess\,sup}$ is the essential supremum of $|\mathbf{u}|$ on Ω [15, 29].

Let $\alpha = (\alpha_1, \dots, \alpha_n)$ be an n -tuple of nonnegative integers α_i . For any nonnegative integer k , the space $C^k(\Omega)$ is the space of functions $\mathbf{u}$ which, together with all

partial derivatives $\partial^\alpha \mathbf{u}$ of orders $|\alpha| \leq k$, are continuous on Ω [15].

For $s \in \mathbb{R}^+$, the L^2 -based Sobolev space $H^s(\Omega)$ is the space of all functions for which the norm is finite [29]. The norm is defined as

$$\|\mathbf{u}\|_{H^s}^2 := (2\pi)^3 \sum_{k \in \mathbb{Z}^3} (1 + |k|^{2s}) |\widehat{\mathbf{u}}_k|^2.$$

In physical space, the norm is defined as

$$\|\mathbf{u}\|_{H^s}^2 = \sum_{0 \leq |\alpha| \leq s} \|\partial^\alpha \mathbf{u}\|_{L^2}^2.$$

For any function $\mathbf{u} \in H^s(\mathbb{T}^3)$ the $\dot{H}^s$ seminorm is defined by

$$\|\mathbf{u}\|_{\dot{H}^s}^2 := (2\pi)^3 \sum_{k \in \mathbb{Z}^3} |k|^{2s} |\widehat{\mathbf{u}}_k|^2.$$

The seminorm in physical space is defined as

$$\|\mathbf{u}\|_{\dot{H}^s}^2 = \sum_{|\alpha|=k} \int_{\Omega} |\partial^\alpha \mathbf{u}|^2 d\mathbf{x}.$$

B Proof of Relation (2.25)

In this Appendix we prove (2.25) which shows that the parameter σ determines the relative 'size' of the Gevrey space in which we search for optimal initial conditions.

Proof. Let $\sigma_1, \sigma_2 \in \mathbb{R}$ be defined such that $0 < \sigma_1 < \sigma_2$. Therefore, from (2.22)

$$\begin{aligned} \|\mathbf{v}\|_{G^{\sigma_2}} &= \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma_2|D|} \mathbf{v} \cdot \mathbf{v} \, d\mathbf{x} \right]^{1/2} \\ &\geq \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma_1|D|} \mathbf{v} \cdot \mathbf{v} \, d\mathbf{x} \right]^{1/2} = \|\mathbf{v}\|_{G^{\sigma_1}} \\ &\geq \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s e^0 \mathbf{v} \cdot \mathbf{v} \, d\mathbf{x} \right]^{1/2} = \|\mathbf{v}\|_{G^0} \\ &= \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s \mathbf{v} \cdot \mathbf{v} \, d\mathbf{x} \right]^{1/2} = \|\mathbf{v}\|_{H^s}. \end{aligned}$$

Thus,

$$\|\mathbf{v}\|_{G^{\sigma_2}} \geq \|\mathbf{v}\|_{G^{\sigma_1}} \geq \|\mathbf{v}\|_{G^0} = \|\mathbf{v}\|_{H^s}.$$

It follows from this that if $\|\mathbf{v}\|_{G^{\sigma_2}} < \infty$ then $\|\mathbf{v}\|_{G^{\sigma_1}} < \infty$ and so if $v \in G^{\sigma_2}$ then $v \in G^{\sigma_1}$. However, the converse is not true. The same argument shows that if $v \in G^{\sigma_1}$ then $v \in G^0$, which is equivalent to $v \in H^s$. Therefore,

$$G^{\sigma_2} \subset G^{\sigma_1} \subset G^0 = H^s.$$

□

C Rescaling the Euler equations

In this Appendix we discuss the scaling symmetry of the Euler-Voigt equation (1.4).

Given a solution $\mathbf{u}$ of the Euler equations (1.2), we construct a new scaled solution with

$$\tau = \lambda t, \quad \text{and} \quad \mathbf{y} = \beta \mathbf{x}.$$

where $\lambda, \beta > 0$ are constants. Our new solution $\mathbf{v}$ is defined as

$$\mathbf{v} = \mathbf{v}(\mathbf{y}, \tau) = \gamma \mathbf{u}(\mathbf{y}/\beta, \tau/\lambda).$$

Therefore

$$\partial_t \mathbf{u} = \frac{1}{\gamma} \frac{\partial \mathbf{v}}{\partial \tau} \frac{d\tau}{dt} = \frac{\lambda}{\gamma} \partial_\tau \mathbf{v}.$$

and

$$\nabla \mathbf{u} = \frac{1}{\gamma} \nabla \mathbf{v} \frac{d\mathbf{y}}{d\mathbf{x}} = \frac{\beta}{\gamma} \nabla \mathbf{v}.$$

Substituting $\mathbf{v}$ into equation (1.2a)

$$\frac{\lambda}{\gamma} \partial_\tau \mathbf{v} + \frac{\beta}{\gamma^2} (\mathbf{v} \cdot \nabla) \mathbf{v} + \nabla q = 0,$$

where q is defined as whatever pressure is required to satisfy the incompressibility condition. This can be rearranged to the form,

$$\frac{\lambda\gamma}{\beta}\partial_\tau\mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} + \nabla q' = 0.$$

Therefore, our function $\mathbf{v}$ can only be a solution to the Euler equations if

$$\frac{\lambda\gamma}{\beta} = 1.$$

For the Taylor Green Vortex with $L = 2\pi$ and $V_0 = 1$, let

$$\gamma = \lambda = \beta = 1.$$

In order to rescale this to the domain with $L = 1$ we must set $\beta = 1/2\pi$. Additionally, we want our rescaled solution to have the same time scale as the original solution and so we must set $\lambda = 1$. Therefore, for our scaled solution $\mathbf{v}$ to be a solution of the Euler equations we must set $\gamma = V_0 = 1/2\pi$.

D Rescaling the Euler-Voigt equations

In this Appendix we discuss the scaling symmetry of the Euler-Voigt equation (1.4).

Given a solution $\mathbf{u}_\alpha$ of the Euler-Voigt equations (1.4), we construct a new scaled solution with

$$\tau = \lambda t, \quad \kappa = \beta \alpha, \quad \mathbf{y} = \beta \mathbf{x}.$$

where $\lambda, \beta > 0$ are constants. The variables $\mathbf{x}$ and α are both scaled by β because they are both spatial and we want to keep the relative size of the filter to the domain size fixed. Our new solution $\mathbf{v}$ is defined as

$$\mathbf{v} = \mathbf{v}(\mathbf{y}, \tau) = \gamma \mathbf{u}_\alpha(\mathbf{y}/\beta, \tau/\lambda).$$

Therefore

$$\partial_t \mathbf{u}_\alpha = \frac{1}{\gamma} \frac{\partial \mathbf{v}}{\partial \tau} \frac{d\tau}{dt} = \frac{\lambda}{\gamma} \partial_\tau \mathbf{v}.$$

and

$$\nabla \mathbf{u}_\alpha = \frac{1}{\gamma} \nabla \mathbf{v} \frac{d\mathbf{y}}{d\mathbf{x}} = \frac{\beta}{\gamma} \nabla \mathbf{v}.$$

and

$$\partial_t \Delta \mathbf{u}_\alpha = \partial_t \frac{\beta^2}{\gamma} \Delta \mathbf{v} = \frac{\lambda \beta^2}{\gamma} \partial_\tau \Delta \mathbf{v}.$$

Substituting $\mathbf{v}$ into equation (1.4a)

$$-\alpha^2 \frac{\lambda \beta^2}{\gamma} \partial_\tau \Delta \mathbf{v} + \frac{\lambda}{\gamma} \partial_\tau \mathbf{v} + \frac{\beta}{\gamma^2} (\mathbf{v} \cdot \nabla) \mathbf{v} + \nabla q = 0,$$

where q is defined as whatever pressure is required to satisfy the incompressibility condition. This can be rearranged to the form,

$$-\kappa^2 \partial_t \Delta \mathbf{v} + \partial_\tau \mathbf{v} + \frac{\beta}{\lambda \gamma} (\mathbf{v} \cdot \nabla) \mathbf{v} + \nabla q' = 0.$$

Therefore, our function $\mathbf{v}$ can only be a solution to the Euler-Voigt equations if

$$\frac{\beta}{\lambda \gamma} = 1.$$

For the Taylor Green Vortex with $L = 2\pi$ and $V_0 = 1$, let $\gamma = \lambda = \beta = 1$. In order to rescale this to the domain with $L = 1$ we must set $\beta = 1/2\pi$. Additionally, we want our rescaled solution to have the same time scale as the original solution and so we must set $\lambda = 1$. Therefore, for our scaled solution $\mathbf{v}$ to be a solution of the Euler-Voigt equations we must set $\gamma = V_0 = 1/2\pi$.

E Evaluation of $\|\boldsymbol{\eta}\|_{\dot{H}^1}$ for the General Taylor Green Vortex

In this Appendix we compute the $\dot{H}^1$ seminorm for the **General** Taylor Green Vortex. The $\dot{H}^1$ seminorm is required for properly scaling the initial conditions of the Euler and Euler-Voigt equations.

The general form of the 3D Taylor Green Vortex on the 3D torus Ω with characteristic length L is given in (2.29). To simplify the notation we drop the subscript *TGV* and substitute in

$$x' = 2\pi x/L, \quad y' = 2\pi y/L, \quad z' = 2\pi z/L.$$

The formula for $\|\boldsymbol{\eta}\|_{\dot{H}^1}$ is

$$\|\boldsymbol{\eta}\|_{\dot{H}^1} = \|\nabla \boldsymbol{\eta}\|_{L^2} = \left[\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \boldsymbol{\eta}_j)^2 d\mathbf{x} \right]^{1/2},$$

where ∂_i is the partial derivative with respect to the i -th component and $\boldsymbol{\eta}_j$ is the j -th component of $\boldsymbol{\eta}$. Because the 3rd component of $\boldsymbol{\eta}$ is zero, the partial derivative

is also zero. This leaves us with six integrals to evaluate.

$$\begin{aligned} \sum_{i,j=1}^3 \int_{\Omega} (\partial_i \boldsymbol{\eta}_j)^2 d\mathbf{x} &= \int_{\Omega} (\partial_1 \boldsymbol{\eta}_1)^2 d\mathbf{x} + \int_{\Omega} (\partial_2 \boldsymbol{\eta}_1)^2 d\mathbf{x} + \int_{\Omega} (\partial_3 \boldsymbol{\eta}_1)^2 d\mathbf{x} \\ &\quad + \int_{\Omega} (\partial_1 \boldsymbol{\eta}_2)^2 d\mathbf{x} + \int_{\Omega} (\partial_2 \boldsymbol{\eta}_2)^2 d\mathbf{x} + \int_{\Omega} (\partial_3 \boldsymbol{\eta}_2)^2 d\mathbf{x}. \end{aligned}$$

Evaluating each partial derivative

$$\begin{aligned} \sum_{i,j=1}^3 \int_{\Omega} (\partial_i \boldsymbol{\eta}_j)^2 d\mathbf{x} &= \int_{\Omega} (2\pi V_0/L \cos(x') \cos(y') \cos(z'))^2 d\mathbf{x} \\ &\quad + \int_{\Omega} (-2\pi V_0/L \sin(x') \sin(y') \cos(z'))^2 d\mathbf{x} \\ &\quad + \int_{\Omega} (-2\pi V_0/L \sin(x') \cos(y') \sin(z'))^2 d\mathbf{x} \\ &\quad + \int_{\Omega} (2\pi V_0/L \sin(x') \sin(y') \cos(z'))^2 d\mathbf{x} \\ &\quad + \int_{\Omega} (-2\pi V_0/L \cos(x') \cos(y') \cos(z'))^2 d\mathbf{x} \\ &\quad + \int_{\Omega} (2\pi V_0/L \cos(x') \sin(y') \sin(z'))^2 d\mathbf{x}, \end{aligned}$$

which simplifies to

$$\begin{aligned} \sum_{i,j=1}^3 \int_{\Omega} (\partial_i \boldsymbol{\eta}_j)^2 d\mathbf{x} &= 4\pi^2 V_0^2 / L^2 \int_{\Omega} \cos^2(x') \cos^2(y') \cos^2(z') d\mathbf{x} \\ &\quad + 4\pi^2 V_0^2 / L^2 \int_{\Omega} \sin^2(x') \sin^2(y') \cos^2(z') d\mathbf{x} \\ &\quad + 4\pi^2 V_0^2 / L^2 \int_{\Omega} \sin^2(x') \cos^2(y') \sin^2(z') d\mathbf{x} \\ &\quad + 4\pi^2 V_0^2 / L^2 \int_{\Omega} \sin^2(x') \sin^2(y') \cos^2(z') d\mathbf{x} \\ &\quad + 4\pi^2 V_0^2 / L^2 \int_{\Omega} \cos^2(x') \cos^2(y') \cos^2(z') d\mathbf{x} \\ &\quad + 4\pi^2 V_0^2 / L^2 \int_{\Omega} \cos^2(x') \sin^2(y') \sin^2(z') d\mathbf{x}. \end{aligned}$$

Because the integrand is separable, the integrals can be reduced to products of,

$$\int_0^L \cos^2(x') dx = \frac{L}{2}, \quad \text{and} \quad \int_0^L \sin^2(x') dx = \frac{L}{2}.$$

Therefore, each of the six integrals is equal to

$$\frac{4\pi^2 V_0^2}{L^2} \frac{L^3}{8}.$$

The sum of the six integrals is

$$\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \boldsymbol{\eta}_j)^2 d\mathbf{x} = 6 \frac{4\pi^2 V_0^2}{L^2} \frac{L^3}{8} = 3\pi^2 V_0^2 L.$$

Substituting into the formula for $\|\cdot\|_{\dot{H}^1}$ gives

$$\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = [3\pi^2 V_0^2 L]^{1/2} = \sqrt{3L}\pi V_0.$$

F Evaluation of $\|\nabla \times \boldsymbol{\eta}\|_{L^\infty}$ for the General Taylor Green Vortex

In this Appendix we compute the L^∞ norm of the vorticity for the General Taylor Green Vortex. The L^∞ norm of the vorticity is required for properly scaling the initial conditions of the Euler and Euler-Voigt equations.

The general form of the 3D Taylor Green Vortex on the 3D torus Ω with characteristic length L is given in (2.29). To simplify the notation we drop the subscript *TGV* and substitute in

$$x' = 2\pi x/L, \quad y' = 2\pi y/L, \quad z' = 2\pi z/L.$$

The vorticity of $\boldsymbol{\eta}$ is

$$\nabla \times \boldsymbol{\eta} = \begin{bmatrix} \partial_2 \eta_3 - \partial_3 \eta_2 \\ \partial_3 \eta_1 - \partial_1 \eta_3 \\ \partial_1 \eta_2 - \partial_2 \eta_1 \end{bmatrix}$$

where ∂_i is the partial derivative with respect to the i -th component and $\boldsymbol{\eta}_j$ is the j -th component of $\boldsymbol{\eta}$. Substituting in the partial derivatives

$$\nabla \times \boldsymbol{\eta} = \begin{bmatrix} 0 - (-(2\pi V_0/L) \cos(x') \sin(y') \sin(z')) \\ -(2\pi V_0/L) \sin(x') \cos(y') \sin(z') - 0 \\ (2\pi V_0/L) \sin(x') \sin(y') \cos(z') + (2\pi V_0) \sin(x') \sin(y') \cos(z'), \end{bmatrix}$$

which simplifies to

$$\nabla \times \boldsymbol{\eta}_{TG} = \begin{bmatrix} (2\pi V_0/L) \cos(x') \sin(y') \sin(z') \\ -(2\pi V_0/L) \sin(x') \cos(y') \sin(z') \\ (4\pi V_0/L) \sin(x') \sin(y') \cos(z') \end{bmatrix}.$$

The $\|\nabla \times \boldsymbol{\eta}\|_{L^\infty}$ norm is the maximum absolute value of the components of $\nabla \times \boldsymbol{\eta}$.

$$\begin{aligned} \|\nabla \times \boldsymbol{\eta}_{TG}\|_{L^\infty} &= \max(|(2\pi V_0/L) \cos(x') \sin(y') \sin(z')| \\ &\quad , |-(2\pi V_0/L) \sin(x') \cos(y') \sin(z')| \\ &\quad , |(4\pi V_0/L) \sin(x') \sin(y') \cos(z')|) \end{aligned}$$

The maximums are

$$\|\nabla \times \boldsymbol{\eta}_{TG}\|_{L^\infty} = \max((2\pi V_0/L), (2\pi V_0/L), (4\pi V_0/L))$$

Therefore,

$$\|\nabla \times \boldsymbol{\eta}_{TG}\|_{L^\infty} = 4\pi V_0/L.$$

G Linearization of the Euler-Voigt System

In this Appendix we derive the linearization of the Euler-Voigt equations. It is used to construct the gradient of the Objective functional.

The Euler-Voigt equations are defined as,

$$\begin{cases} -\alpha^2 \partial_t \Delta \mathbf{u}_\alpha + \partial_t \mathbf{u}_\alpha + (\mathbf{u}_\alpha \cdot \nabla) \mathbf{u}_\alpha + \nabla p_\alpha = 0 \\ \nabla \cdot \mathbf{u}_\alpha = 0 \\ \mathbf{u}_\alpha|_{t=0} = \boldsymbol{\eta}_\alpha. \end{cases}$$

Substituting in $\mathbf{u}_\alpha = \mathbf{u}_\alpha + \mathbf{u}'_\alpha$, $\boldsymbol{\eta}_\alpha = \boldsymbol{\eta}_\alpha + \boldsymbol{\eta}'_\alpha$, and $p_\alpha = p_\alpha + p'_\alpha$,

$$\begin{cases} -\alpha^2 \partial_t \Delta (\mathbf{u}_\alpha + \mathbf{u}'_\alpha) + \partial_t (\mathbf{u}_\alpha + \mathbf{u}'_\alpha) + ((\mathbf{u}_\alpha + \mathbf{u}'_\alpha) \cdot \nabla) (\mathbf{u}_\alpha + \mathbf{u}'_\alpha) + \nabla (p_\alpha + p'_\alpha) = 0 \\ \nabla \cdot (\mathbf{u}_\alpha + \mathbf{u}'_\alpha) = 0 \\ (\mathbf{u}_\alpha + \mathbf{u}'_\alpha)|_{t=0} = \boldsymbol{\eta}_\alpha + \boldsymbol{\eta}'_\alpha. \end{cases}$$

By expanding each term we obtain,

$$\begin{cases} -\alpha^2 \partial_t \Delta \mathbf{u}_\alpha - \alpha^2 \partial_t \Delta \mathbf{u}'_\alpha + \partial_t \mathbf{u}_\alpha + \partial_t \mathbf{u}'_\alpha + \mathbf{u}_\alpha \cdot \nabla \mathbf{u}_\alpha + \\ \mathbf{u}_\alpha \cdot \nabla \mathbf{u}'_\alpha + \mathbf{u}'_\alpha \cdot \nabla \mathbf{u}_\alpha + \mathbf{u}'_\alpha \cdot \nabla \mathbf{u}'_\alpha + \nabla p_\alpha + \nabla p'_\alpha = 0 \\ \nabla \cdot (\mathbf{u}_\alpha + \mathbf{u}'_\alpha) = 0 \\ (\mathbf{u}_\alpha + \mathbf{u}'_\alpha)|_{t=0} = \boldsymbol{\eta}_\alpha + \boldsymbol{\eta}'_\alpha. \end{cases}$$

Substituting in the Euler-Voigt equations we simplify to

$$\begin{cases} -\alpha^2 \partial_t \Delta \mathbf{u}'_\alpha + \partial_t \mathbf{u}'_\alpha + \mathbf{u}_\alpha \cdot \nabla \mathbf{u}'_\alpha + \mathbf{u}'_\alpha \cdot \nabla \mathbf{u}_\alpha + \mathbf{u}'_\alpha \cdot \nabla \mathbf{u}'_\alpha + \nabla p'_\alpha = 0 \\ \nabla \cdot \mathbf{u}'_\alpha = 0 \\ \mathbf{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'_\alpha. \end{cases}$$

Linearizing the system by removing non-linear terms, we derive the linearized Euler-Voigt system

$$\begin{cases} -\alpha^2 \partial_t \Delta \mathbf{u}'_\alpha + \partial_t \mathbf{u}'_\alpha + \mathbf{u}_\alpha \cdot \nabla \mathbf{u}'_\alpha + \mathbf{u}'_\alpha \cdot \nabla \mathbf{u}_\alpha + \nabla p'_\alpha = 0 \\ \nabla \cdot \mathbf{u}'_\alpha = 0 \\ \mathbf{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'_\alpha. \end{cases}$$

This can be rearranged to the form

$$\begin{cases} \partial_t \mathbf{u}'_\alpha + (I - \alpha^2 \Delta)^{-1} (\mathbf{u}_\alpha \cdot \nabla \mathbf{u}'_\alpha + \mathbf{u}'_\alpha \cdot \nabla \mathbf{u}_\alpha + \nabla p'_\alpha) = 0 \\ \nabla \cdot \mathbf{u}'_\alpha = 0 \\ \mathbf{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'_\alpha. \end{cases}$$

H Derivation of the Adjoint System

In this Appendix we derive the adjoint system (3.38) using the duality-pairing relation (3.39). We drop the subscript α from $\mathbf{u}_\alpha$, $\mathbf{u}_\alpha^*$, and $\mathbf{u}'_\alpha$ to simplify the notation. Additionally, ~~we substitute the integration with the following notation.~~

$$\int_0^T \int_\Omega \mathbf{u} \, d\mathbf{x} \, dt = \int_D \mathbf{u} \, d\mathbf{x} \, dt.$$

We begin by expanding (3.39) into ~~smaller~~ integrals which will be ~~solved~~ separately.

$$\left(\mathcal{L} \begin{bmatrix} \mathbf{u}' \\ p' \end{bmatrix}, \begin{bmatrix} \mathbf{u}^* \\ p^* \end{bmatrix} \right) = \int_D \partial_t \mathbf{u}' \cdot \mathbf{u}^* \, d\mathbf{x} \, dt \tag{H.64a}$$

$$+ \int_D (I - \alpha^2 \Delta)^{-1} (\mathbf{u} \cdot \nabla \mathbf{u}') \cdot \mathbf{u}^* \, d\mathbf{x} \, dt \tag{H.64b}$$

$$+ \int_D (I - \alpha^2 \Delta)^{-1} (\mathbf{u}' \cdot \nabla \mathbf{u}) \cdot \mathbf{u}^* \, d\mathbf{x} \, dt \tag{H.64c}$$

$$+ \int_D \nabla p' \cdot \mathbf{u}^* \, d\mathbf{x} \, dt \tag{H.64d}$$

$$+ \int_D (\nabla \cdot \mathbf{u}') p^* \, d\mathbf{x} \, dt \tag{H.64e}$$

Integrating (H.64a) in time with integration by parts, where $u = \mathbf{u}^*$ and $dv = \partial_t \mathbf{u}'$.

$$\int_D \partial_t \mathbf{u}' \cdot \mathbf{u}^* d\mathbf{x} dt = \int_{\Omega} \mathbf{u}^* \cdot \mathbf{u}'|_{t=0}^{t=T} d\mathbf{x} - \int_D \partial_t \mathbf{u}^* \cdot \mathbf{u}' d\mathbf{x} dt.$$

To solve (H.64b) we first split the integrand into its components, and use einstein notation to simplify.

$$\int_D (I - \alpha^2 \Delta)^{-1} (\mathbf{u} \cdot \nabla \mathbf{u}') \cdot \mathbf{u}^* d\mathbf{x} dt = \int_D (I - \alpha^2 \Delta)^{-1} u_i \partial_i u'_j u_j^* d\mathbf{x} dt.$$

Integrating (H.64b) in space with integration by parts, where $u = (I - \alpha^2 \Delta)^{-1} u_i u_j^*$ and $dv = \partial_i u'_j$.



$$= \int_0^T (I - \alpha^2 \Delta)^{-1} u_i u_j^* u'_j|_{\partial\Omega} dt - \int_D u'_j \partial_i ((I - \alpha^2 \Delta)^{-1} u_i u_j^*) d\mathbf{x} dt.$$

Because Ω has periodic boundaries, the first integral is zero, and we can expand the second integral.

$$= \int_D u'_j [u_i \partial_i ((I - \alpha^2 \Delta)^{-1} u_i) - (I - \alpha^2 \Delta)^{-1} u_j^* \partial_i u_i] d\mathbf{x} dt.$$

The flow is divergence free and so $\partial_i u_i = 0$. We can convert the second integral back to vector notation to obtain

$$\int_D (I - \alpha^2 \Delta)^{-1} (\mathbf{u} \cdot \nabla \mathbf{u}') \cdot \mathbf{u}^* d\mathbf{x} dt = - \int_D (\mathbf{u} \cdot \nabla (I - \alpha^2 \Delta)^{-1} \mathbf{u}^*) \cdot \mathbf{u}' d\mathbf{x} dt.$$

To solve (H.64c) we first split the integrand into its components, and use einstein notation to simplify.

$$\int_D (I - \alpha^2 \Delta)^{-1} (\mathbf{u} \cdot \nabla \mathbf{u}') \cdot \mathbf{u}^* d\mathbf{x} dt = \int_D (I - \alpha^2 \Delta)^{-1} u'_i (\partial_i u_j) u_j^* d\mathbf{x} dt.$$

The flow is divergence free and so we can convert express this as

$$= \int_D (I - \alpha^2 \Delta)^{-1} \partial_i (u'_i u_j) u_j^* d\mathbf{x} dt.$$

Integrating (H.64c) in space with integration by parts, where $u = (I - \alpha^2 \Delta)^{-1} u_j^*$ and $dv = \partial_i (u'_i u_j)$.

$$= \int_0^T (I - \alpha^2 \Delta)^{-1} u_j^* u'_i u_j dt - \int_D u'_i u_j \partial_i (I - \alpha^2 \Delta)^{-1} u_j^* d\mathbf{x} dt$$

Because Ω has periodic boundaries, the first integral is zero. We can convert the second integral back to vector notation to obtain

$$\int_D (I - \alpha^2 \Delta)^{-1} (\mathbf{u} \cdot \nabla \mathbf{u}') \cdot \mathbf{u}^* d\mathbf{x} dt = - \int_D (\nabla (I - \alpha^2 \Delta)^{-1} \mathbf{u}^*)^\top \mathbf{u} \cdot \mathbf{u}' d\mathbf{x} dt.$$

To solve (H.64d) we first split the integrand into its components, and use einstein notation to simplify.

$$\int_D \nabla p' \cdot \mathbf{u}^* d\mathbf{x} dt = \int_D \partial_i p' u_i^* d\mathbf{x} dt.$$

Integrating (H.64d) in space with integration by parts, where $u = u_i^*$ and $dv = \partial_i p'$.

$$\int_D \nabla p' \cdot \mathbf{u}^* d\mathbf{x} dt = \int_0^T \mathbf{u}^* p'|_{\partial\Omega} dt - \int_D p' \partial_i u_i^* d\mathbf{x} dt.$$

Because Ω has periodic boundaries, the first integral is zero, and we can convert the second integral back to vector notation to obtain

$$\int_D \nabla p' \cdot \mathbf{u}^* d\mathbf{x} dt = - \int_D p' (\nabla \cdot \mathbf{u}^*) d\mathbf{x} dt.$$

To solve (H.64e) we first split the integrand into its components, and use einstein notation to simplify.

$$\int_D (\nabla \cdot \mathbf{u}') p^* d\mathbf{x} dt = \int_D (\partial_i u'_i) p^* d\mathbf{x} dt$$

Integrating (H.64e) in space with integration by parts, where $u = p^*$ and $dv = \partial_i u'_i$.

$$\int_D (\nabla \cdot \mathbf{u}') p^* d\mathbf{x} dt = \int_0^T p^* \mathbf{u}'|_{\partial\Omega} dt - \int_D u'_i \partial_i p^* d\mathbf{x} dt.$$

Because Ω has periodic boundaries, the first integral is zero, and we can convert the second integral back to vector notation to obtain

$$\int_D (\nabla \cdot \mathbf{u}') p^* d\mathbf{x} dt = - \int_D \mathbf{u}' \cdot \nabla p^* d\mathbf{x} dt.$$

Combining the integrals, we can express the LHS of (3.39) as

$$\left(\mathcal{L} \begin{bmatrix} \mathbf{u}' \\ p' \end{bmatrix}, \begin{bmatrix} \mathbf{u}^* \\ p^* \end{bmatrix} \right) = \int_{\Omega} \mathbf{u}^* \cdot \mathbf{u}'|_{t=0}^{t=T} d\mathbf{x} - \int_D \mathbf{u}' \cdot \partial_t \mathbf{u}^* d\mathbf{x} dt. \quad (\text{H.65a})$$

$$- \int_D (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}^* \cdot \mathbf{u}) \cdot \mathbf{u}' d\mathbf{x} dt. \quad (\text{H.65b})$$

$$- \int_D (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}^*)^\top \mathbf{u} \cdot \mathbf{u}' d\mathbf{x} dt \quad (\text{H.65c})$$

$$- \int_D p' (\nabla \cdot \mathbf{u}^*) d\mathbf{x} dt \quad (\text{H.65d})$$

$$- \int_D \mathbf{u}' \cdot \nabla p^* d\mathbf{x} dt. \quad (\text{H.65e})$$

Using this and the RHS of (3.39) we obtain the adjoint system,

$$\mathcal{L}^* \begin{bmatrix} \mathbf{u}_\alpha^* \\ p_\alpha^* \end{bmatrix} := \begin{bmatrix} -\partial_t \mathbf{u}_\alpha^* - (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}_\alpha^* + (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}_\alpha^*)^T) \mathbf{u}_\alpha - \nabla p_\alpha^* \\ -\nabla \cdot \mathbf{u}_\alpha^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

To derive the terminal condition of the adjoint system we use the RHS of (3.39).

$$\int_{\Omega} \mathbf{u}_\alpha^*(T) \cdot \mathbf{u}'_\alpha(T) d\mathbf{x} = \int_{\Omega} \mathbf{u}_\alpha^*(0) \cdot \mathbf{u}'_\alpha(0) d\mathbf{x}.$$

From the two forms of the Gâteaux differential, (3.36) and (3.40), we obtain

$$\int_{\Omega} \mathbf{u}_\alpha^*(T) \cdot \mathbf{u}'_\alpha(T) d\mathbf{x} = 2 \int_{\Omega} |D| \mathbf{u}_\alpha^*(T) \cdot |D| \mathbf{u}'_\alpha(T) d\mathbf{x}.$$

Therefore,

$$\mathbf{u}_\alpha^*(T) = 2|D|^2 \mathbf{u}_\alpha(T).$$

I Derivation of the Gâteaux Directional Differential of the Objective Functional

In this Appendix we derive **an expression for** the Gâteaux directional differential of the objective functional, $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha)$ in terms of the $\dot{H}^1$ inner product and L^2 inner product. This differential is used to **obtain** the gradient of the objective functional.

We begin by substituting the second definition of (3.32) into (3.34) to obtain

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\|\mathbf{u}_\alpha(T; \boldsymbol{\eta}_\alpha + \epsilon \boldsymbol{\eta}'_\alpha)\|_{\dot{H}^1}^2 - \|\mathbf{u}_\alpha(T; \boldsymbol{\eta}_\alpha)\|_{\dot{H}^1}^2].$$

Using the $\dot{H}^1$ seminorm as defined in Appedix A, and taking the limit $\epsilon \rightarrow 0$, we can express $\Phi'_{\alpha,T}$ in the form of the $\dot{H}^1$ inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = 2 \langle \mathbf{u}_\alpha(T), \mathbf{u}'_\alpha(T) \rangle_{\dot{H}^1}.$$

Using the integral form of the $\dot{H}^s$ seminorm from Appendix A, the Gâteaux differential can be expressed in physical space as

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \int_{\mathbb{T}^3} 2|D|^2 \mathbf{u}_\alpha(\mathbf{x}, T) \cdot \mathbf{u}'_\alpha(\mathbf{x}, T) d\mathbf{x}.$$

We now make use of the **terminal condition in the** adjoint system terminal condition imposed in Appendix H

$$\mathbf{u}_\alpha^*(x, T) = 2|D|^2 \mathbf{u}_\alpha(\mathbf{x}, T).$$

Therefore,

$$\Phi'_{\alpha, T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \int_{\mathbb{T}^3} \mathbf{u}_\alpha^*(\mathbf{x}, T) \cdot \mathbf{u}'_\alpha(\mathbf{x}, T) d\mathbf{x}.$$

Now using the duality-pairing 3.39

$$\Phi'_{\alpha, T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \int_{\mathbb{T}^3} \mathbf{u}_\alpha^*(\mathbf{x}, 0) \cdot \boldsymbol{\eta}'_\alpha(\mathbf{x}) d\mathbf{x},$$

which lets us express the value of $\Phi'_{\alpha, T}$ with the L^2 inner product

$$\Phi'_{\alpha, T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \langle \mathbf{u}_\alpha^*(0), \boldsymbol{\eta}'_\alpha \rangle_{L^2}.$$

Therefore,

$$\Phi'_{\alpha, T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = 2 \langle \mathbf{u}_\alpha(T), \mathbf{u}'_\alpha(T) \rangle_{\dot{H}^1} = \langle \mathbf{u}_\alpha^*(0), \boldsymbol{\eta}'_\alpha \rangle_{L^2}.$$

As described in Section 3.3, the differential can also be expressed as a Gevrey inner product using the Riesz representation theorem (3.35). Using the definition of the Gevrey inner product (2.22), the L^2 inner product can be converted to the Gevrey inner product,

$$\langle \mathbf{u}_\alpha^*(0), \mathbf{u}'_\alpha(0) \rangle_{L^2} = \langle (1 + |D|^2)^{-3} e^{-2\sigma|D|} \mathbf{u}_\alpha^*(0), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma}.$$

Therefore, we have two forms of the Gâteaux differential as a Gevrey inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) = \langle \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma} = \langle (1 + |D|^2)^{-3} e^{-2\sigma|D|} \boldsymbol{u}_\alpha^*(0), \boldsymbol{\eta}'_\alpha \rangle_{G^\sigma}.$$

From this we obtain the gradient of the objective functional

$$\nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) = (1 + |D|^2)^{-3} e^{-2\sigma|D|} \boldsymbol{u}_\alpha^*(0).$$

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